

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines

Uritect: An Edge-Computing Smartphone System for Urinalysis
Risk Profiling via Illumination-Adaptive Colorimetry and
Bayesian Late Fusion

An Undergraduate Thesis
Presented to the Faculty of the
College of Information and Communications Technology
West Visayas State University
La Paz, Iloilo City

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Computer Science

by
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Karlo Roel F. Montenegro
Steven Ken E. Pontillas

May 2026

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines

Approval Sheet

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CHAPTER 1 INTRODUCTION TO THE STUDY

Background of the Study and Conceptual Framework

Urinalysis remains one of the most fundamental and widely utilized diagnostic procedures in primary healthcare, serving as a rapid and cost-effective method for detecting urinary tract infections (UTIs), renal abnormalities, metabolic irregularities, and other systemic conditions. Standard reagent dipsticks enable simultaneous semi-quantitative assessment of multiple biochemical parameters—including nitrite, leukocyte esterase, protein, glucose, blood, ketones, pH, and specific gravity—within minutes. These parameters provide clinically useful indicators of urinary tract conditions and kidney-related abnormalities. Due to its low cost and minimal infrastructure requirements, dipstick urinalysis is extensively used in community health centers and rural primary care facilities.

Despite its accessibility, dipstick interpretation in many healthcare environments still relies on manual visual comparison of reagent pad color changes with printed reference charts. This process introduces variability arising from ambient lighting conditions, reader

subjectivity, fatigue, and inconsistencies in timing during reagent development. In resource-constrained healthcare environments such as Philippine Rural Health Units (RHUs), where automated reflectance photometry analyzers may be unavailable due to financial and infrastructure limitations, manual interpretation remains a common workflow [1], [2]. Previous studies have documented measurable inter-reader variability in visual dipstick analysis [3], [4], particularly under inconsistent illumination conditions such as variations in warm and cool fluorescent lighting [5], [6]. These operational constraints may reduce reproducibility and affect the reliability of early disease screening in routine clinical practice.

Recent developments in mobile health (mHealth) technologies have explored the use of smartphone cameras as accessible tools for colorimetric analysis of urine reagent strips. Controlled laboratory investigations have shown that smartphone-based image analysis can achieve promising classification performance across several urinalysis parameters [5], [7]. However, a persistent technical challenge in these systems involves sensitivity to illumination conditions. Variations in color temperature,

light intensity, camera sensor characteristics, and automatic exposure adjustments may introduce non-linear color distortions that affect classification stability. Many existing approaches assume controlled lighting environments or rely on external calibration accessories, which may limit their practicality in real-world rural or community healthcare settings where illumination conditions are often unpredictable.

In addition to illumination variability, another limitation observed in many smartphone-based urinalysis systems is their focus on unimodal image classification. These systems typically treat dipstick analysis purely as a computer vision task without incorporating relevant clinical context. In routine clinical practice, however, urinalysis findings are rarely interpreted in isolation. Healthcare workers commonly consider patient-reported symptoms and relationships among multiple dipstick parameters when evaluating potential conditions. For example, the simultaneous presence of nitrite and leukocyte esterase may increase the likelihood of bacterial urinary tract infection, particularly when accompanied by symptoms such as painful urination, urinary urgency, or fever [8], [9]. Because individual dipstick markers exhibit different

diagnostic characteristics—for instance, nitrite tests generally demonstrate high specificity but moderate sensitivity [8], [9]—integrating dipstick findings with symptom information may provide a more structured basis for preliminary risk assessment.

These two operational challenges—(1) variability in dipstick color interpretation under inconsistent illumination and (2) the limited integration of symptom information in automated urinalysis tools—highlight opportunities for improving smartphone-assisted screening systems. Addressing both aspects within a unified framework may support healthcare workers in obtaining more consistent dipstick interpretations while also providing structured decision-support information during preliminary screening.

In response to these challenges, this study proposes the development of a smartphone-based Clinical Decision Support System (CDSS) designed to assist urinalysis screening in resource-constrained healthcare environments. The system incorporates illumination normalization techniques to improve the stability of smartphone-based dipstick color classification under varying lighting conditions. Image-based classification outputs are then combined with structured symptom inputs using a

probabilistic reasoning framework to estimate risk levels for selected urinary conditions, including urinary tract infection indicators and renal-related abnormalities.

The proposed application is implemented using an edge computing architecture, allowing image processing, classification, and probabilistic reasoning to operate directly on the smartphone device without requiring internet connectivity. This design supports deployment in Rural Health Units and other primary healthcare environments where network access may be limited. The system functions strictly as a screening-oriented decision support tool intended to assist healthcare personnel in interpreting urinalysis findings. It does not replace laboratory diagnostic procedures or professional medical evaluation.

By combining illumination-aware smartphone image analysis with symptom-informed probabilistic reasoning, this study aims to demonstrate the feasibility of a context-appropriate mobile screening support system for urinalysis in resource-limited healthcare settings.

Objectives of the Study

This study aims to design, develop, and evaluate an offline, edge-computing smartphone application for automated urinalysis screening that integrates illumination-adaptive computer vision with a multimodal Bayesian Late Fusion framework for disease risk estimation.

Specifically, it aims to:

1. Construct and develop a calibrated dataset of urinalysis reagent strip images using a combined dataset of volunteer urine specimens and standardized synthetic control solutions imaged under three controlled lighting conditions (2700K, 4000K, and 5500K) for classifier training and algorithm calibration.
2. Develop an Adaptive White Balance (AWB) normalization method that compensates for color temperature variations by referencing the unreacted white plastic region of the strip as a static calibration point.
3. Develop an offline mobile application featuring an in-app automated Region of Interest (ROI) segmentation method using adaptive thresholding and contour

detection to accurately isolate the ten reagent pads under variable backgrounds.

4. Train and evaluate a per-analyte k-Nearest Neighbors (k-NN) classification model on the constructed dataset—assessing training performance using accuracy, macro F1-score, sensitivity, specificity, and Cohen's Kappa per analyte parameter—and integrate this visual classifier into a Bayesian Late Fusion framework to estimate posterior risk probabilities for urinary tract infection (UTI) and renal abnormality indications.
5. Evaluate the application's usability, functionality, and performance using established software quality metrics based on the ISO/IEC 25010 standard.

Significance of the Study

The findings and technological outputs of this study are expected to contribute to the fields of health informatics, computer vision, and public healthcare delivery. By developing an offline, multimodal smartphone-based Clinical Decision Support System (CDSS) for urinalysis screening, this research offers both practical and academic value to multiple stakeholders.

Medical Technologists and Rural Health Workers

In resource-constrained primary care facilities, healthcare workers often manage high patient volumes without access to automated laboratory analyzers. The proposed system functions as an objective digital decision-support tool that may reduce inter-reader variability associated with inconsistent lighting conditions, subjective color perception, and visual fatigue. By combining illumination-adaptive image processing with Bayesian-based disease risk estimation, the system provides structured probabilistic outputs to assist healthcare workers in conducting more consistent preliminary screening

while preserving clinical judgment for final interpretation.

Patients in Geographically Isolated and Disadvantaged Areas (GIDAs)

Patients in rural communities frequently encounter geographic and financial barriers to accessing laboratory diagnostics. This offline mobile application supports screening accessibility by enabling digitally assisted urinalysis interpretation within local health centers or guided monitoring contexts. Through the integration of dipstick colorimetric findings and structured symptom inputs using a probabilistic Bayesian aggregation framework, the system generates context-aware risk stratification for conditions such as urinary tract infections and basic renal impairment indicators. This may support earlier identification of at-risk individuals and facilitate timely referral when necessary.

Local Government Units (LGUs) and the Department of Health (DOH)

From a public health perspective, the study introduces a minimal-infrastructure diagnostic support solution that operates entirely on-device without requiring internet

connectivity or cloud-based computation. By leveraging an edge computing architecture, local government units may deploy the system on standard Android smartphones, potentially strengthening screening consistency in Rural Health Units (RHUs) without substantial additional infrastructure costs.

The Field of Computer Science and Health Informatics

Academically, this research contributes a hybrid computational framework that integrates illumination-adaptive computer vision using k-Nearest Neighbors (k-NN) with Bayesian Late Fusion for multimodal clinical reasoning. The study demonstrates the feasibility of implementing probabilistic posterior risk estimation entirely offline on resource-limited mobile devices. This hybrid architecture may inform future research in smartphone-assisted colorimetric diagnostics and other point-of-care testing systems where multimodal data fusion is required.

Delimitation of the Study

To ensure feasibility, methodological focus, ethical compliance, and successful completion within the undergraduate timeframe, the researchers establish the following deliberate boundaries for the system's design, development, and evaluation:

1. Data Development and Validation Scope

Algorithm calibration, controlled testing, and validation are conducted using both standardized synthetic urine control solutions and human urine samples applied to standard 10-parameter reagent dipsticks. The training dataset combines two sources: volunteer urine specimens processed by licensed medical technologists at the partner RHU, and standardized synthetic control solutions – specifically Level 1 Negative and Level 2 Positive standardized preparations – imaged under the same three controlled lighting conditions. Together, these constitute the combined reference dataset used for k-NN classifier training, AWB calibration, and Bayesian Late Fusion parameterization. A separate validation phase evaluates agreement between system-generated interpretations and manual interpretations performed by licensed healthcare

personnel at the partner facility under real-world conditions. These samples are collected and handled exclusively by licensed professionals as part of routine clinical procedures. The researchers do not directly collect, store, or transport biological specimens. The study does not constitute a clinical diagnostic trial, does not establish population-level diagnostic performance, and does not evaluate patient outcomes or clinical management decisions.

2. Platform and Hardware Scope

The mobile application is developed exclusively for the Android operating system and is primarily evaluated on devices running Android 14. To maintain a consistent baseline for on-device edge computing performance, the application is strictly architected for compatibility with entry-level Android smartphones. System development and empirical validation are constrained to devices operating on a minimum of 4GB of Random Access Memory (RAM) to ensure accessibility without compromising algorithm execution. Cross-platform deployment, including compatibility with iOS devices, is beyond the scope of this study.

The visual classification component is calibrated specifically for standard 10-parameter urine reagent dipsticks. Alternative strip configurations (such as 2-parameter or 4-parameter strips) and laboratory-based liquid reagent urinalysis methods are not considered in this research.

3. Environmental and Illumination Scope

The Adaptive White Balance (AWB) normalization algorithm is designed to improve classification stability under ambient indoor lighting conditions commonly encountered in rural clinical settings. The algorithm is calibrated to operate within color temperature ranges between approximately 2700K (warm white) and 5500K (cool daylight).

Lighting environments outside this range—such as extremely low-light conditions requiring heavy camera flash usage or direct outdoor sunlight that may produce strong glare on the wet plastic surface of the dipstick—are not evaluated in this study.

4. Clinical Reasoning Scope

Although the computer vision module extracts colorimetric information from all ten reagent pads and displays the complete semiquantitative results for all ten parameters, the probabilistic disease-level reasoning model is limited to structured risk estimation using four designated Sentinel Analytes: Leukocyte Esterase, Nitrite, Protein, and Glucose. This parameter delimitation is consistent with standard clinical laboratory practice, wherein these four analytes serve as the primary biochemical indicators for the two target screening categories of this study – urinary tract infection (UTI) and renal-related abnormality indicators. The remaining six parameters – Urobilinogen, pH, Blood, Specific Gravity, Ketone, and Bilirubin – are extracted, classified, and displayed in full within the application interface for review by licensed healthcare personnel, but are not incorporated into the Bayesian Late Fusion reasoning engine. These supplemental parameters are intended to support the clinical judgment of the attending medical technologist or physician rather than contribute to the automated risk stratification output.

The symptom integration module therefore includes only predefined inputs relevant to the UTI and renal screening

categories. The system does not attempt comprehensive diagnosis of complex systemic diseases such as advanced hepatic disorders or metabolic conditions.

All outputs generated by the system represent preliminary risk stratification intended to assist healthcare personnel and guided users. These outputs do not replace laboratory confirmation or physician diagnosis.

5. End-User and Deployment Scope

The application is designed primarily as a Clinical Decision Support System (CDSS) for trained healthcare personnel operating in Rural Health Units (RHUs), including medical technologists and Barangay Health Workers (BHWs). The system is intended to support routine urinalysis screening workflows by assisting personnel in interpreting reagent dipstick results using automated image analysis and probabilistic risk estimation.

The system assumes that trained personnel perform proper dipstick handling, timing of reagent reactions, and image capture procedures in accordance with standard urinalysis protocols. The application does not attempt to verify or enforce procedural compliance during testing.

The system functions solely as a preliminary screening and decision-support tool within clinical or community healthcare settings. It does not prescribe medication and does not provide definitive medical diagnosis. All generated outputs represent risk stratification results that require confirmation through standard laboratory testing and professional medical evaluation.

6. Validation and Evaluation Scope

To evaluate the reliability of the proposed system, the dipstick classification outputs generated by the smartphone application will be compared with manual interpretations performed by trained healthcare personnel using standard reagent strip reference charts. Manual interpretation serves as the reference standard in this study because it reflects the routine urinalysis workflow in many Rural Health Units (RHUs) where automated urinalysis analyzers may not be readily available. This comparison assesses the level of agreement between the system-generated results and conventional visual dipstick interpretation.

Definition of Terms

To ensure clarity and uniformity in the interpretation of this study, the following terms are operationally defined based on how they are utilized within the proposed system:

Adaptive White-Balance (AWB) In this study, Adaptive White Balance refers to the image preprocessing technique used to reduce the effects of color temperature variation in smartphone-captured images. Operationally, the system applies a reference-based gray world assumption by detecting the unreacted white plastic region of the urine reagent strip and using it as a calibration reference. This normalization step adjusts the color balance of the captured image to minimize lighting-induced color shifts before further color feature extraction is performed.

Bayesian Late Fusion Model In the context of this study, this term refers to the probabilistic reasoning component used to estimate disease risk by integrating outputs from dipstick color classification with structured symptom inputs. The system applies Bayesian inference to combine visual classification probabilities and symptom-based likelihoods with predefined prior probabilities for

selected urinary conditions. This probabilistic aggregation produces posterior risk estimates intended to support preliminary screening and does not represent a definitive medical diagnosis.

Clinical Decision Support System (CDSS) In the context of this research, CDSS refers to the offline Android smartphone application being developed. It is operationally defined as a presumptive screening and triage support tool designed to assist Rural Health Unit (RHU) personnel and guided users, and not as a definitive diagnostic replacement for laboratory analyzers or physicians.

Control Solutions (Synthetic Urine) This refers to commercially prepared, in vitro standardized liquids with manufacturer-specified biochemical concentration ranges (e.g., Level 1 Negative and Level 2 Positive controls). Operationally, these solutions are used in place of real human urine during the initial dataset development phase to establish controlled reference labels for algorithm calibration.

Edge Computing This refers to the software architecture of the proposed system. Operationally, it means that all image processing, machine learning classification, and

probabilistic reasoning are executed locally on the smartphone's internal processor (CPU), requiring no reliance on cloud servers, external databases, or active internet connectivity.

HSV Color Space (Hue, Saturation, Value) Unlike the standard RGB (Red, Green, Blue) format, HSV is the mathematical color representation used by the system to quantify dipstick pad colors. Operationally, separating Hue and Saturation from Value reduces dependence on brightness intensity compared to raw RGB analysis, improving robustness under varying lighting conditions.

Illumination-Invariant In this study, this term refers to the system's engineered ability to reduce classification variability caused by lighting-induced color shifts. Operationally, it is evaluated based on maintaining consistent classification performance across predefined lighting conditions (2700K, 4000K, and 5500K) within acceptable variance thresholds.

k-Nearest Neighbors (k-NN) This is the supervised machine learning algorithm used as the system's visual classifier. Operationally, it plots extracted HSV feature values of a scanned dipstick pad within a multidimensional feature

space, computes the distance to the nearest labeled data points in the reference dataset, and outputs a class prediction, with probability estimated from the proportion of nearest neighbors belonging to each class.

Multimodal Classification This refers to the system's ability to process and integrate heterogeneous data types. Operationally, it combines continuous visual features (HSV values extracted from the smartphone camera) with categorical clinical metadata (e.g., Yes/No symptom inputs such as fever or painful urination) to produce structured disease risk estimation.

Reagent Dipstick (10-Parameter) This refers to the physical diagnostic tool used in the study. Operationally, it is a commercially available plastic strip containing ten chemically treated pads (e.g., Leukocytes, Nitrite, Urobilinogen, Protein, pH, Blood, Specific Gravity, Ketone, Bilirubin, Glucose) that undergo color changes upon exposure to synthetic control solutions or urine samples.

Region of Interest (ROI) Segmentation This refers to the computer vision preprocessing step executed prior to color analysis. Operationally, it uses adaptive thresholding and contour detection to locate the dipstick in the captured

image, reduce background interference, and isolate the relevant pixel regions corresponding to the ten reagent pads.

CHAPTER 2 REVIEW OF RELATED STUDIES

Review of Existing and Related Studies

This chapter establishes the academic and technical foundation for the proposed smartphone-based Clinical Decision Support System for urinalysis screening in Philippine Rural Health Units. The review is organized into three parts. The first covers current systems, examining existing automated urinalysis analyzers used in clinical settings and their limitations in resource-constrained environments. The second covers related systems and solutions, surveying both commercial smartphone-based urinalysis platforms and academic prototypes to identify persistent gaps in offline capability, standard dipstick compatibility, and clinical decision support integration. The third presents related studies, which grounds each technical module of the proposed system – illumination calibration, segmentation, feature extraction, multimodal Bayesian reasoning, validation methodology, dataset preparation, colorimetric stability, and regulatory

compliance – in published literature. Together, these three parts justify the design decisions of the proposed system and position it within the broader landscape of mobile diagnostic technology.

Current Systems

Manual Urinalysis Workflow in Philippine Rural Health Units

In Philippine Rural Health Units (RHUs), routine urinalysis is primarily performed using standard 10-parameter reagent dipsticks such as Multistix 10 SG, SD UroColor, or similar cost-effective alternatives, and results are interpreted by manual visual comparison against manufacturer-provided reference color charts [[1]. This workflow remains the dominant diagnostic approach in primary care settings due to limited access to laboratory automation, budget constraints, and infrastructure limitations [2], [3]. These constraints often necessitate the use of shelf-stable, manual point-of-care testing (POCT) over more complex automated systems [4].

Medical technologists in RHUs frequently process high daily testing volumes under documented staffing shortages [5]. Under these operational conditions, dipstick

interpretation is primarily conducted by visually matching reagent pad color changes to printed reference color charts. These assessments typically occur in uncontrolled ambient environments where lighting ranges from warm incandescent to cool daylight illumination. Research shows that differences in lighting color, brightness, and unwanted color tints from the surroundings can noticeably change how the reagent pad colors appear in an image. Combined with inconsistent viewing angles, these factors make visual reading unreliable, especially for parameters whose color changes rely heavily on shades of red [6].

Beyond illumination-related variability, manual workflows are also susceptible to kinetic and timing errors. Each reagent pad requires a specific incubation period; manufacturers generally recommend reading results between 60 and 120 seconds after dipping, with leukocyte esterase being particularly sensitive to over-incubation [7]. In high-volume RHU environments, batch reading practices where multiple strips are processed simultaneously may result in over-incubation or premature interpretation, thereby compromising analytical accuracy. Furthermore, the tropical climate of the Philippines presents additional pre-analytical challenges. Elevated

humidity levels can accelerate the degradation of reagent pads once containers are opened, particularly in facilities lacking consistent environmental control [8].

Manual dipstick reading is inherently dependent on the operator's visual acuity, training, and fatigue level. Reported inter-reader variability ranges from approximately 15-25%, particularly during high workload conditions [9]. Comparative studies on inter-brand dipstick agreement demonstrate that kappa values across key parameters such as bilirubin fall within the range of $\kappa = 0.44\text{--}0.54$, with some brands showing minimal to no agreement ($\kappa \leq 0.33$), indicating that even standardized semi-quantitative instruments produce inconsistent baseline readings across manufacturers [10]. This variability is compounded when results are interpreted visually rather than instrumentally, further undermining the reliability of manual dipstick reading as a consistent diagnostic baseline in high-volume RHU workflows.

Supply-side readiness assessments further indicate that many primary care facilities lack automated analyzers due to high acquisition costs (approximately ₱1-2 million per unit), ongoing maintenance requirements, and unstable power infrastructure [11]. Consequently, RHUs rely

predominantly on manual interpretation despite their documented susceptibility to environmental, kinetic, and human error.

Automated Urinalysis Systems in Higher-Level Facilities

In contrast, tertiary hospitals and urban diagnostic laboratories commonly employ automated urinalysis analyzers such as the CLINITEK Novus [12]. These systems utilize reflectance photometry to quantify colorimetric changes by measuring the intensity of light reflected at specific wavelengths relative to a calibrated reference light source [13].

Technically, these devices standardize the environment by using Light Emitting Diodes (LEDs) at precise target wavelengths matched to the primary chromogenic changes on the reagent pads. This eliminates the unpredictability of ambient SPD. These systems enable high-throughput processing of up to 240 tests per hour [14]. By eliminating operator-dependent visual estimation and controlling environmental variables, these systems achieve superior reproducibility and inter-test consistency.

Despite their advantages, automated systems are often inaccessible to rural health facilities due to high costs

and infrastructure gaps. These platforms require expensive equipment, specialized maintenance, and proprietary supplies that are rarely available in geographically isolated areas. Furthermore, inconsistent power in remote provinces prevents the steady use of sensitive laboratory tools [11]. As a result, advanced diagnostic technology remains concentrated in cities, leaving rural health units dependent on manual methods.

Technological Gap

The current diagnostic landscape reveals a structural imbalance. Rural facilities depend on subjective manual interpretation methods vulnerable to environmental illumination shifts, reagent timing variability, and operator-dependent inconsistency. Conversely, urban institutions benefit from standardized optical quantification systems that minimize these sources of variability but require substantial financial and infrastructural investment.

This gap remains a practical problem: the technology for accurate, lighting-controlled color analysis exists, but it is out of reach for most rural health facilities. What is needed is a solution that can deliver consistent

and objective results using equipment already available in these settings specifically, the cameras found in modern smartphones without requiring the infrastructure or resources that rural facilities typically lack.

Related Systems or Solutions

Commercial Smartphone-Based Urinalysis Applications

Several commercial smartphone-based urinalysis solutions have been developed to support home-based urine testing and wellness monitoring. While these platforms demonstrate the technical feasibility of mobile colorimetric analysis, their system architectures and operational models differ substantially from the workflow requirements of Philippine RHUs.

Vivoo is a consumer-oriented urine testing application that utilizes proprietary dipsticks to measure selected wellness parameters including ketones, pH, hydration, Vitamin C, Magnesium, Calcium, Sodium, and Oxidative Stress [15]. The platform employs smartphone-based image capture and automated color interpretation; however, it operates within a closed consumable ecosystem requiring manufacturer-specific test strips and subscription-linked

services. This design introduces a vendor-dependent procurement model, commonly referred to as vendor lock-in, which limits interoperability with widely used clinical dipsticks in Philippine public health facilities, such as Multistix 10 SG. Furthermore, Vivoo is primarily positioned as a lifestyle and wellness monitoring tool rather than a regulated clinical diagnostic system. The proprietary and non-transparent nature of its analytical algorithms restricts independent validation and local calibration—an important consideration for clinical deployment in regulated healthcare environments.

Another commercially validated solution is Dip.io, developed by Healthy.io. Dip.io is an FDA-cleared smartphone-based urinalysis system intended for home testing and remote clinical supervision [16], [17]. The platform enables multi-parameter dipstick analysis using smartphone imaging combined with a proprietary physical calibration board to standardize color interpretation. Despite its clinical validation and cross-platform compatibility across both iOS and Android devices, Dip.io relies on cloud-based processing for interpretation and result transmission. In geographically isolated and disadvantaged areas, where internet connectivity may be

intermittent or absent, this cloud dependency constrains operational reliability.

SpectraPhone is a compact, smartphone-powered spectrometer designed to provide quantitative, clinical-grade urinalysis for early chronic kidney disease (CKD) detection. By utilizing custom acrylic light pipes and a diffraction grating, the system captures 2,051 distinct wavelengths across the 400–750 nm range, achieving a high resolution of 0.17 nm. The device significantly outperforms traditional dipstick testing, which is often limited by qualitative ambiguity and low resolution. In empirical validation using an iPhone 13 Pro Max, SpectraPhone demonstrated high precision in quantifying CKD biomarkers, achieving an R^2 of 0.9913 for hematuria, 0.9981 for standard albuminuria, and up to 0.9997 for reagent-enhanced albuminuria using bromophenol blue. Its low production cost—ranging from \$20 to \$90 USD, and compatibility with mainstream smartphones make it a viable solution for point-of-care diagnostics in both at-home and resource-limited settings [18].

Moreover, most commercial smartphone-based urinalysis platforms focus on parameter detection and categorical reporting rather than probabilistic disease modeling. While

multi-parameter outputs are displayed, these systems typically do not compute disease-specific posterior probabilities using integrated statistical frameworks. As a result, users receive descriptive biochemical results without structured risk stratification or uncertainty quantification tailored to primary care clinical decision-making.

Academic Smartphone-Based Urinalysis Prototypes

In parallel with commercial development, multiple academic research groups have proposed smartphone-based urinalysis prototypes leveraging computer vision and machine learning to enhance colorimetric accuracy. Balbach et al. developed a system incorporating illumination-invariant preprocessing through RGB-to-HSV color space transformation and reference-based calibration to mitigate lighting variability [19]. Thakur et al. applied supervised machine learning models to classify reagent strip pad colors under varying illumination conditions, reporting performance degradation when environmental control was reduced [20].

Similarly, Kim et al. developed an artificial intelligence-based framework using a k-nearest neighbor (k-NN) classification algorithm validated against a standard automated urine analyzer [21]. Their system achieved high diagnostic performance, reporting 98.7% accuracy for nitrites and 97.3% for glucose, while identifying 75 seconds as the optimal incubation period for image capture. It is worth noting, however, that this study employed custom novel reagent strips rather than standard clinical dipsticks such as Multistix 10 SG—a distinction relevant to RHU deployment, where consumable standardization is a key procurement consideration. Additionally, the study was characterized by controlled laboratory conditions, including a standardized smartphone model (Samsung Galaxy S9) and a fixed stand to maintain a consistent capture distance and angle. Real-world deployment in Philippine RHUs introduces additional complexities, particularly cross-device generalization challenges. Variations in CMOS sensor characteristics, automatic white balancing algorithms, and proprietary image signal processing pipelines across low- to mid-range Android devices—commonly used in Philippine public health settings—may significantly influence the performance of such k-NN models.

The Doughnut-shaped Nearness Urine Tester (DONUT) is an academic prototype designed to achieve illumination invariance without the need for external calibration cards or controlled lighting boxes. The system employs a novel reagent strip geometry where each central test pad is surrounded by a ring of reference color swatches, ensuring that both the sample and the reference are subjected to the exact same spectral power distribution (SPD) and intensity. Utilizing a smartphone-based algorithm that performs circular Hough transforms for pad detection and analyzes color differences in the CIE Lab* and HSV color spaces, the DONUT system preserves the relative differences between colors even as ambient light varies from warm (~2700 K) to cool (~5500 K). This prototype provides a critical precedent for the present work, proving that spatial nearness and relative colorimetric matching can effectively mitigate the subjective variability of uncontrolled environments [22].

Many academic prototypes are also trained and validated using relatively small, homogeneous datasets collected under controlled experimental conditions. This introduces potential dataset bias and limits external validity when models are deployed across diverse patient

populations, lighting environments, and device hardware configurations [6]. Without multi-center validation or heterogeneous data sampling, which accounts for variations in CMOS sensor characteristics, model generalization in real-world rural healthcare settings remains uncertain.

Furthermore, while academic studies demonstrate technical feasibility, there is limited discussion to regulatory compliance pathways, clinical governance frameworks, or post-market surveillance considerations. Transitioning from a research prototype to a deployable medical decision support tool (MDST) requires alignment with diagnostic device regulations, including Software as a Medical Device (SaMD) classification frameworks as defined by the International Medical Device Regulators Forum [23] and applicable data privacy laws [24]. These areas, alongside institutional approval processes for GIDAs, remain frequently underexplored in research-oriented implementations [25].

TABLE I SYSTEM COMPARISON

Study	Offline Operation	No Hardware Attachment	Standard Strip Support	Symptom Data Input	Disease Risk Output	RHU Deployment Ready
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Proposed System	✓	✓	✓	✓	✓	✓
Vivoo [15]	X	X	X	X	X	X
Dip.io [17]	X	X	✓	X	X	X
SpectraPhone [18]	✓	X	X	X	X	X
DONUT [22]	✓	X	X	X	X	X
Kim et al. [21]	X	X	X	X	X	X

Table I compares seven key features across existing commercial and academic smartphone-based urinalysis systems. The features were selected to highlight the most critical requirements for deployment in Philippine Rural Health Units: purpose alignment, target setting, hardware dependency, offline capability, symptom integration, risk score output, and standard dipstick compatibility. Among all reviewed systems, the proposed system is the only one that satisfies all seven criteria simultaneously. Existing commercial platforms such as Vivoo and Dip.io are designed primarily for home and consumer use, require proprietary hardware or calibration accessories, and depend on cloud

connectivity. Academic prototypes such as DONUT and Kim et al. demonstrate technical rigor but were developed under controlled laboratory conditions and do not integrate clinical symptoms or produce probabilistic risk outputs. No existing system combines offline operation, attachment-free design, standard dipstick compatibility, symptom-based reasoning, and explainable risk stratification within a single platform. This gap directly motivates the design of the proposed system and positions it as a contextually appropriate solution for frontline urinalysis screening in resource-limited primary care settings.

Synthesis of Related Solutions

A review of both commercial and academic systems reveals a recurring gap. Commercial platforms are designed with consumer usability and regulatory compliance in mind, but tend to be costly, dependent on internet connectivity, and incompatible with standard clinical dipsticks used in primary care settings. Academic prototypes, while technically rigorous, are typically evaluated under controlled laboratory conditions and rarely address practical deployment concerns such as varying device capabilities, limited processing power, or integration into existing clinical workflows.

Beyond these individual limitations, neither category consistently combines dipstick-based biochemical analysis with patient-reported symptoms to support clinical decision-making. Most existing systems treat urinalysis as a standalone diagnostic test rather than one component of a broader clinical assessment, which reduces their utility for frontline health workers who must make decisions with limited resources and no specialist support.

These observations highlight the need for a middle-ground solution, one that incorporates research-grade illumination correction and on-device machine learning within a practical, offline-capable Android application purpose-built for deployment in Philippine Rural Health Units.

Related Studies

This section presents the theoretical and empirical foundations underlying each processing module of the proposed system. The literature is organized to mirror the system's core technical pipeline – beginning with Illumination Calibration, Segmentation, Feature Extraction, and Multimodal Analysis – followed by subsections on

Validation, Explainability, and Ethics; Dataset Acquisition and Reagent Preparation; Colorimetric Stability and Single-Capture Protocol Justification; and Regulatory and Standards Context. By anchoring each module and design decision in established academic research, this study ensures that the selected image processing, machine learning, and clinical reasoning techniques are mathematically sound, clinically justified, and appropriate for deployment in the Philippine Rural Health Unit (RHU) environment.

Illumination Calibration

Color Constancy and the Gray World Assumption

The digital interpretation of colorimetric diagnostics is fundamentally challenged by the variability of ambient illumination. The human visual system overcomes this challenge through color constancy, a perceptual mechanism by which consistent colors are perceived despite shifts in the illumination spectrum [26]. The computational analog to this mechanism is the Gray World Assumption, originally formalized by Buchsbaum [27] and comprehensively surveyed by Gijsenij et al. [28], which posits that the average

reflectance of a scene approximates a neutral gray. While these global statistical assumptions are foundational, recent advancements by Yan et al. [29] have refined this principle by shifting from global scene averages to local reflectance differences. This modern framework demonstrates that traditional assumptions like Gray World remain technically viable in complex, multi-illuminant environments when integrated into hybrid models that account for local spatial variations [29].

The proposed system incorporates a reference-based modification of this assumption. Rather than computing the global scene mean, the mean intensity calculation is restricted to a known "True White" reference region of the dipstick pad. This targeted approach achieves more reliable chromatic adaptation, as it isolates the reference from the chromatic content of the reagent pads themselves [19]. The approach is consistent with findings by Finlayson and Trezzi [30], who demonstrated that spatially restricted gray-world computations outperform global averaging when the scene contains dominant chromatic regions. This principle has been further refined in recent region-weighted frameworks, such as that proposed by Wang et al. [31], which utilizes multi-scale semantic guidance to

suppress the influence of biased chromatic regions and prioritize locally stable neutral patches for illuminant estimation.

Von Kries Chromatic Adaptation and Gain Scaling

The gain-scaling transformation applied by the Illumination Calibrator is grounded in the Von Kries Diagonal Model of Chromatic Adaptation, as formalized by the International Commission on Illumination [26]. This model posits that chromatic adaptation can be achieved by independently scaling the gain of each primary sensor channel (R, G, and B) through a diagonal matrix transformation, rather than through full-rank matrix inversion. Each channel gain is defined as the ratio between a global target intensity and the measured mean intensity of the corresponding channel within the reference region.

This formulation is computationally advantageous because diagonal scaling requires only three scalar multiplications per pixel, making it well-suited for pixel-wise transformation on mobile central processing units (CPUs) operating under battery and thermal constraints

[19]. The model operates on the assumption that the sensor channels respond independently to illumination changes, a condition that holds sufficiently well for consumer-grade mobile camera sensors under the illumination range of 2700K to 5500K targeted by this system.

Several illumination correction strategies have been evaluated for suitability in attachment-free smartphone colorimetric diagnostics. Willmott et al. (2020) demonstrated that ambient subtraction using flash/no-flash image pairs achieves device-independent colorimetric consistency, but this approach requires two sequential captures and custom per-device calibration, creating workflow friction in point-of-care settings [32]. Machine learning-based color constancy approaches achieve state-of-the-art performance on natural scenes but introduce non-transparent transformations, require pre-training on large datasets, and impose computational overhead incompatible with offline edge deployment [33].

The reference-based white balance modification adopted in this study avoids these failure modes by restricting the illuminant estimate to the physically known neutral plastic substrate of the dipstick, rather than inferring it from unconstrained scene statistics. This spatially restricted

reference approach is consistent with the color correction principle validated by Balbach et al. (2021), who confirmed that embedding a known-neutral reference within the image provides reproducible illumination compensation across diverse smartphone hardware and ambient lighting conditions without requiring hardware attachments [19].

TABLE II ILLUMINATION COMPARISON

Study / Algorithm	No Hardware Required	Low Computational Cost	Handles Biased Scenes	POC Viable
Gijssenij et al. [28] – Gray World	✓	✓	X	X
Gijssenij et al. [28] – White Patch	✓	✓	X	X
Gijssenij et al. [28] – Gamut Mapping	X	X	✓	X
Nixon et al. [32] – Ambient Subtraction	✓	✓	✓	X
Domislović et al. [33]– ML-Based	✓	X	✓	X

Balbach et al. [19] – Reference-Based AWB	✓	✓	✓	✓
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Table II compares six illumination correction methods across the criteria most relevant to this study's point-of-care deployment context. Only the reference-based AWB approach – as validated by Balbach et al. [19] and adopted in this study – satisfies all four criteria simultaneously, distinguishing it from scene-statistic methods that fail under chromatically biased conditions and ML-based approaches that exceed the computational constraints of budget Android hardware.

Segmentation

Static Geometry Mapping over Deep Learning Object Detection

Image segmentation in point-of-care diagnostics requires accurate localization of reagent pad regions of interest (ROIs) without imposing prohibitive computational demands on the host device. Contemporary object detection frameworks, such as You Only Look Once (YOLO) and its derivatives, achieve high generalization across diverse visual scenes but require substantial computational resources, including floating-point operations, GPU

acceleration, and significant memory bandwidth [34]. These requirements are incompatible with the processing capacity and battery life of budget-specification Android devices prevalent in resource-constrained Philippine RHU deployments.

As an alternative, the proposed system employs static geometry mapping—a technique grounded in the prior structural knowledge of the dipstick's fixed physical layout. This approach is consistent with the segmentation methodology employed by Ra et al., who demonstrated that geometric constraint-based ROI partitioning—wherein the center of each reagent pad region is computed from fixed structural coordinates and ROI boundaries are derived from those centroids—achieves reliable pad localization for standardized ten-parameter urine test strips while remaining computationally feasible on mobile platforms [22]. By exploiting the invariant geometry of the dipstick rather than learning it from data, the system eliminates the need for a large annotated segmentation dataset and reduces the risk of localization failure in low-texture or high-glare capture conditions [35].

Grid Slicing and ROI Partitioning

The segmentation algorithm divides a vertically oriented dipstick cutout into exactly ten equally spaced slots corresponding to the ten reagent pads of a standard ten-parameter urinalysis strip. This grid slicing approach is based on the calculated usable height of the image region, from which uniform slot boundaries are derived through linear partitioning. The center of each slot is then localized using the midpoint formula, yielding the centroid coordinates used for color sampling in the subsequent Feature Extraction module.

This method is consistent with prior implementations of smartphone-based dipstick readers, which have demonstrated that fixed-geometry partitioning achieves reliable pad localization when image capture is guided by an on-screen alignment overlay [36]. The reliability of this approach is contingent on controlled capture alignment, which the system enforces through the user interface.

Feature Extraction

Cylindrical Color Modeling and HSV Color Space Conversion

The feature extraction module transforms the sampled pixel data from each reagent pad ROI into numerical feature

vectors suitable for machine learning classification. This transformation begins with a color space conversion from the Red-Green-Blue (RGB) representation produced by the camera sensor to the Hue-Saturation-Value (HSV) cylindrical color model.

The selection of the HSV color space for clinical colorimetry is supported by its capacity to decouple chromatic information from luminance. In the RGB model, all three channels are simultaneously affected by changes in illumination intensity, making it difficult to isolate chemically induced color changes from environmental interference [37]. This limitation is particularly consequential in multi-analyte strip analysis, where subtle pad-to-pad colorimetric differences must be resolved independently [38]. In contrast, the HSV model encodes chromatic information primarily in the Hue channel, which remains relatively invariant to changes in illumination intensity and surface shadow [19]. This property is clinically significant because the colorimetric reactions on urinalysis reagent pads produce shifts in hue—such as yellow-to-green in protein detection or tan-to-purple in leukocyte esterase detection—that are more reliably captured by the Hue channel than by any single RGB

component. While perceptually uniform color spaces such as CIELAB offer superior colorimetric precision and are considered the gold standard in quantitative colorimetry, HSV was selected for this system due to its native support in the OpenCV for Android library and its lower computational overhead, which is critical for real-time processing on entry-level mobile hardware [39].

Statistical Averaging and Variance Reduction

Each reagent pad ROI contains a distribution of pixel values that reflects both the underlying chemical reaction and sensor-level noise sources, including photon shot noise and thermal noise. To produce a stable feature value from this distribution, the system computes the mean Hue (μH) and mean Saturation (μS) across all pixels within the ROI. This statistical averaging approach implements a variance reduction technique established in digital image processing, whereby the mean of a large pixel sample converges toward the true signal value as individual noise contributions cancel out [37].

The incorporation of the Saturation channel alongside Hue is supported by Yang et al., who demonstrated through RGB-to-CIE Lab color space transformation on commercial

urinalysis strips that human-vision-based models capture subtle colorimetric shifts in low-concentration analytes—including glucose and protein—more effectively than luminance-weighted RGB models, enabling high accuracy in point-of-care settings [40].

Integrative Evidence Analysis

Pathophysiology and Clinical Correlates

The classification logic of the integrative analysis module is grounded in the established pathophysiology of the three target conditions screened by the system.

Urinary Tract Infection (UTI) is primarily characterized by the colonization of the urinary tract by pathogenic bacteria, most commonly *Escherichia coli*. This triggers an inflammatory response in the urothelium, resulting in the recruitment of neutrophils (pyuria) and occasional micro-hematuria [7]. The system's visual evidence stream for UTI incorporates the Griess Reaction, in which bacterial nitrate reductase converts dietary nitrate to nitrite, producing a detectable color change on the nitrite reagent pad. The leukocyte esterase (LE) pad simultaneously detects the enzymatic activity of infiltrating neutrophils. Clinical symptoms such as dysuria

and urinary frequency serve as corroborating evidence in the system's probabilistic framework [41].

Renal conditions, including Chronic Kidney Disease (CKD) and Glomerulonephritis, involve structural damage to the glomerular filtration barrier, which permits the pathological passage of albumin into the urine [42]. The detection of proteinuria in the proposed system is based on the protein error of indicators principle: the reagent pad contains tetrabromophenol blue buffered at $\text{pH} \approx 3.0$, which shifts from yellow to green or blue in the presence of albumin due to the release of hydrogen ions [8]. Clinical symptoms such as peripheral edema and fatigue are integrated into the system's weighting logic to flag persistent proteinuria for urgent renal triage.

k-Nearest Neighbors Classification

The visual classification component of the Late Fusion Analysis module is implemented using the k-Nearest Neighbors (k-NN) algorithm. k-NN is a non-parametric, instance-based learning method that classifies a query sample by identifying the k most similar examples in a labeled reference dataset and assigning the majority class among those neighbors [43], [44]. Similarity is measured

using the Euclidean distance between the query's feature vector and each reference vector in the two-dimensional (μH , μS) space. The validity of the Euclidean metric in this context is contingent on isotropic feature scaling, a condition satisfied by normalizing the HSV feature vectors prior to classification.

The k-NN algorithm was selected over parametric alternatives—such as logistic regression or naïve bayes classifiers and over deep learning architectures for two reasons. First, k-NN does not require a large training dataset to define a decision boundary, making it suitable for the moderate sample sizes attainable in a Philippine RHU clinical validation study [45]. Second, k-NN provides instance-based explanations by revealing which specific training neighbors contributed to a given prediction, a core principle of Explainable AI (XAI) that is discussed further [46].

The performance of k-NN is governed by the bias-variance tradeoff: small values of k produce low-bias but high-variance decision boundaries sensitive to noise, while large values of k produce smoother boundaries at the cost of increased bias. The selection of an optimal k is

addressed empirically through cross-validation in the system's evaluation phase [43].

The selection of k-NN over alternative supervised classifiers was informed by the specific constraints of this study's deployment context. Kim et al. (2025) developed an AI-based program using a k-NN classification algorithm trained on 2,612 real hospital urine test strip images, achieving 98.7% accuracy for nitrite and 97.3% for glucose detection, demonstrating the clinical viability of k-NN for smartphone-based colorimetric urinalysis [21]. Kim and Cho (2022) further demonstrated that RGB semiquantitative analysis combined with machine learning achieves reliable colorimetric classification performance under variable environmental conditions, supporting the use of color-space features for dipstick grade level discrimination [47].

Support Vector Machines offer strong generalization by identifying optimal separating hyperplanes, but their default hard-label outputs and the additional calibration burden of probabilistic extensions via Platt scaling reduce transparency in clinical settings. Random Forests require larger training sets for reliable ensemble variance. Decision trees overfit with small, geometrically

constrained datasets such as the controlled HSV color clusters expected in reagent pad colorimetry.

By contrast, k-NN directly leverages the geometric regularity of color space: semi-quantitative dipstick grades produce distinct color clusters, and nearest-neighbor retrieval is conceptually equivalent to the visual matching task performed by trained healthcare workers. This alignment between algorithm behavior and the clinical tasks combined with the absence of a training phase and support for soft probability estimation via neighbor proportions, supports k-NN as the most appropriate classifier for this deployment context. Flaucher et al. (2022) demonstrated that Hue-channel-based color comparison achieves the highest classification performance ($F1 = 0.81$) among colorimetric matching strategies for smartphone urinalysis in home environments, supporting the use of Hue as the primary feature dimension in the present system's k-NN classifier [48].

TABLE III CLASSIFIER COMPARISON

Study	Low Data	Prob. Output	Interpretable	Small Dataset
Hastie et al. [43] – SVM	X	X	X	✓

Hastie et al. [43] – Random Forest	X	✓	X	X
Hastie et al. [43] – Decision Tree	✓	X	✓	X
Barredo Arrieta et al. [46] – Deep Learning	X	✓	X	X
Kim et al. [47] – k-NN	✓	✓	✓	✓
Flaucher et al. [48]– CNN + Hue Distance	X	X	X	X

Table III compares six supervised classification algorithms across the criteria most relevant to this study's deployment context. Among all evaluated algorithms, only k-NN – as supported by Kim et al. [21] – satisfies all four criteria simultaneously, reinforcing its selection as the classification algorithm for the proposed system. Deep learning and ensemble methods such as Random Forest are excluded due to their high data demands and lack of transparency, while CNN-based approaches such as Flaucher et al. [48] demonstrate limited suitability for small-dataset, interpretable classification tasks.

Bayesian Fusion and Clinical Reasoning

Relying exclusively on the k-NN visual classifier would render the system a digital color reader—a tool that assigns diagnostic classes based solely on the mathematical similarity of HSV vectors to reference data, without accounting for clinical symptomatology. Such a unimodal approach is clinically inadequate because it treats visual results (continuous colorimetric data) and patient symptoms (categorical clinical evidence) as equivalent coordinates in a mathematical space, failing to represent their differing diagnostic specificity and sensitivity [49].

Clinical diagnosis is inherently a probabilistic, iterative process where every piece of evidence—whether a chemical biomarker or a physical symptom—updates the posterior probability of a diagnosis. To implement this effectively in a computational environment, the system must utilize a structured information fusion architecture [50].

Information fusion architectures are broadly classified into three paradigms based on the stage at which different data streams are integrated [51].

Early Fusion (Feature-Level): This approach concatenates raw features (e.g., continuous HSV vectors and

binary categorical symptoms) into a single feature space before classification. While it allows for early correlation detection, it is unsuitable here because combining geometrically distinct data types produces "meaningless distances" in the feature space, reducing classifier reliability [51].

Intermediate Fusion (Representation-Level): Modality-specific representations are merged at hidden layers, typically within deep learning models. While powerful for cross-modal interactions, this "black box" approach is incompatible with Explainable AI (XAI) requirements, as users cannot trace per-modality contributions [49].

Late Fusion (Decision-Level): Each data stream produces independent outputs combined at the final decision stage. This is optimal for heterogeneous inputs like visual and symptom data, preserving interpretability while handling complementary errors [52].

The proposed system utilizes a Bayesian Late-Fusion architecture, treating the k-NN classification result as probabilistic visual evidence that is subsequently updated by patient-reported symptoms via Bayesian inference [50]. This dual-stream integration addresses the inherent

limitations of chemical reagent strips; for instance, clinical symptoms like dysuria serve as a weighting factor to increase the posterior probability of a UTI, effectively mitigating false-negative results from nitrite-negative Gram-positive infections or inadequate bladder incubation [53]

To ensure clinical relevance, baseline "prior probabilities" are anchored to Philippine epidemiological data. The system integrates hospital morbidity rates from the DOH Field Health Services Information System (FHSIS) to establish a UTI prevalence baseline[54]. Furthermore, renal risk estimates are informed by Philippine Society of Nephrology data, which indicates that CKD affects approximately 11.2% of the national population .[55] Grounding the framework in these local statistics prevents the over-diagnosis risks associated with generic international datasets.

By separating visual perception from probabilistic clinical reasoning, the system functions as a transparent Clinical Decision Support System (CDSS). This architecture ensures that any risk classification can be traced back to its individual source—whether the dipstick image, reported symptoms, or both—upholding the Explainable AI (XAI)

principles required for deployment in primary care settings [46].

Validation, Explainability, and Ethics

Diagnostic Accuracy and Cross-Validation

The statistical reliability of the proposed system is governed by established diagnostic accuracy study protocols. Validating a point-of-care tool requires direct comparison against a laboratory gold standard—in this case, automated benchtop urinalysis or urine culture. The primary performance metrics reported are sensitivity, specificity, Precision, Recall, and the F1-score, derived from a confusion matrix analysis [56]. While overall accuracy provides a general overview, it is a misleading metric in imbalanced medical datasets where the prevalence of positive cases is low. The F1-score, which harmonizes Precision and Recall, is therefore the primary metric for evaluating per-condition diagnostic performance.

To mitigate overfitting in small to moderate sample sizes, the study employs stratified k-fold cross-validation, reporting performance across multiple folds rather than a single train-test split. This approach reduces variance in estimated sensitivity and specificity

and aligns with best practices in medical machine learning validation [45], [57]. The minimum sample size for each condition is determined following the method described by Hajian-Tilaki, which derives sample size estimates from target sensitivity and specificity values with specified confidence intervals [45]. The Receiver Operating Characteristic (ROC) Curve and the Area Under the Curve (AUC) are additionally reported to characterize system performance across varying decision thresholds [56].

Explainable AI and Transparent Decision-Making

In the deployment of intelligent systems within healthcare, the tradeoff between model complexity and interpretability is a critical ethical and practical consideration [46]. This study incorporates Explainable AI (XAI) principles—as taxonomized by Barredo Arrieta et al.—by selecting k-NN over deep learning architectures [46]. This instance-based transparency fosters clinician trust and ensures that the system functions as a decision support tool rather than an autonomous diagnostic authority, consistent with the CDSS design principles described by Shortliffe and Cimino [50].

Edge Computing and Data Privacy

The decision to implement on-device classification using a local SQLite database is supported by edge computing principles in healthcare [58]. Processing medical data locally eliminates the need to transmit sensitive patient images over public networks, directly addressing the data privacy requirements of the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). This local-first architecture also ensures operational continuity in Geographically Isolated and Disadvantaged Areas (GIDA), where cloud connectivity is unreliable or absent—a known failure point in cloud-dependent diagnostic tools [58]. Recent implementations of on-device medical AI demonstrate that local processing achieves significant latency reduction while maintaining diagnostic privacy, making it a viable and ethically sound framework for rural health equity.

Dataset Acquisition and Reagent Preparation

Contemporary smartphone-based urinalysis systems employ diverse methodological strategies for training dataset acquisition, broadly categorized across three primary validation paradigms. Synthetic and spiked-solution approaches utilize artificial urine matrices with precisely controlled analyte concentrations to establish reproducible

ground-truth labels across semi-quantitative grade boundaries (Sarigul et al., 2019). Sarigul et al. demonstrated that artificial urine protocols comprising physiologically representative electrolyte and metabolite concentrations yield reproducible results, with spectroscopic validation contexts supporting dipstick applications [59].

Commercial bilinear control solutions represent a second validation paradigm, employing manufacturer-validated negative (Level 1) and positive (Level 2) liquid controls to bracket the clinical measurement range [60]. These ready-to-use controls provide reference standards for analyzer calibration, with assigned values across multiple analytes reported by manufacturers (Tru Laboratories). Unlike synthetic research formulations designed for experimental flexibility, commercial controls offer standardized matrix compositions optimized for quality assurance [60].

Residual clinical specimen analysis constitutes the third primary approach, utilizing anonymized leftover samples from routine laboratory workflows to introduce authentic matrix variability including endogenous interferences [61], [62]. Olivetti et al. demonstrated that

residual clinical specimens processed under Institutional Review Board oversight provide ground-truth labels comparable to reference analytical methods [61]. Pohanka and Zakova further established that smartphone camera-derived colorimetric measurements align with spectrophotometric reference values across clinically relevant concentration ranges [62].

Methodological synthesis combines commercial controls with clinical specimens to mitigate systematic measurement biases attributable to urine composition effects [61]. Standardization protocols enhance cross-study comparability, with illumination-controlled imaging (warm incandescent ~2700 K, neutral fluorescent ~4000 K, cool daylight ~5500 K) and consistent dip-and-read timing (2-second immersion, 60-second reaction) serving as consensus preanalytical standards [8], [22], [63].

Colorimetric Stability of Reagent Pads and Single-Capture Protocol Justification

The URS-10T Reagent Strip for Urinalysis instructs users to read all parameters within 60 seconds after dipping, with the exception of leukocytes, which may be read up to 120 seconds [14]. Color changes appearing after

2 minutes are considered to have no diagnostic significance as specified in the product insert of the URS-10T Reagent Strip used in this study [64]. These timepoints indicate the earliest and latest intervals during which each reagent pad's color change is considered valid for visual interpretation.

When a camera is used instead of the human eye, the interpretation of these timing windows differs slightly. The recommended reading windows indicate when the color reaction has sufficiently developed for interpretation; they do not imply that the color disappears or rapidly changes immediately afterward. For most parameters, the color stabilizes once the chemical reaction is complete and remains relatively consistent throughout the 120-second observation period [7], [64]. This behavior is characteristic of dry reagent pad chemistry: once the reaction reaches completion, the resulting color typically remains stable within the normal testing timeframe [8].

Several smartphone-based urinalysis studies have adopted a similar approach. Kim and Cho captured a single image of the dipstick and extracted RGB color values to classify all parameters without requiring images at multiple time points [47]. Noor Azhar and colleagues also utilized a single

image captured at a fixed time after dipping and reported reliable detection across all ten parameters [6]. These findings support the feasibility of using a single image captured after sufficient reaction time to evaluate the entire reagent strip.

Based on these observations, the system in this study captures a single image at 120 seconds after the strip is dipped. By this time, all ten parameters have had sufficient time to fully develop their color reactions, including leukocytes, which require the longest development period. This approach eliminates the need for users to capture multiple images at different time points, which may be impractical in busy Rural Health Unit (RHU) settings. Furthermore, it mirrors the operational principle of automated urinalysis analyzers, which internally read each pad at its optimal reaction time before producing a single consolidated result [65]. The difference in this implementation is that a smartphone camera is used instead of an integrated optical sensor.

Regulatory and Standards Context

Before deploying any software that supports clinical decisions, it is important to understand the regulatory

frameworks that govern how such tools are classified, evaluated, and used. This section identifies the key standards relevant to the proposed system and explains how they inform its design boundaries and validation requirements.

Software as a Medical Device (SaMD)

The International Medical Device Regulators Forum (IMDRF) defines SaMD as any software that serves a medical purpose while running on a general-purpose device rather than dedicated medical hardware [23]. The proposed system fits this definition – it runs on an ordinary Android smartphone and produces risk assessments intended to guide clinical decisions at the point of care.

The IMDRF further classifies SaMD by risk, based on two factors: how seriously the output could affect a patient, and how directly it drives a clinical action [66]. Because the proposed system is designed to *screen and flag* rather than to *diagnose or treat*, it falls under a lower-risk category – one that informs a healthcare worker's judgment rather than replacing it. This distinction is important: it means the system is not held to the same

evidentiary burden as a diagnostic device, which is appropriate given the study's scope.

At the national level, the Philippine FDA has released a draft circular requiring AI-based medical software to follow the IMDRF SaMD classification framework, using the ASEAN Medical Device Directive as its basis for risk categorization [67]. This aligns with how the proposed system is already positioned in this study.

Clinical Accuracy Standards

Two international standards define the accuracy benchmarks relevant to this system. The CLSI guideline GP16-A3 establishes the procedural and interpretive standards for semi-quantitative urine dipstick analysis, including the acceptable color grade ranges for each analyte [68]. These thresholds serve as the reference against which the system's k-NN classification outputs are compared. ISO 15189:2022 sets quality and competence requirements for medical laboratories and point-of-care testing, including the principle that any diagnostic or screening system must be validated against a recognized reference method – not against manual user interpretation alone [69]. The present study's validation approach

compares system outputs against structured dual-expert manual consensus reading rather than an automated reference analyzer. This is directly attributable to the operational context of the study: primary care facilities in the Philippines, particularly outside urban centers, are ill-equipped to provide diagnostic services due to logistically-hampered distribution of diagnostic equipment and inadequate infrastructure to house and maintain automated analyzers [2]. In the absence of an automated reference standard, dual trained readers with consensus adjudication represents the most operationally feasible reference method available, and its use is consistent with STARD 2015 reporting principles, which require authors to transparently justify reference standard selection based on intended use and practical constraints of the deployment setting [70]. Accordingly, performance metrics in this study should be interpreted as measures of agreement with structured expert consensus rather than absolute diagnostic accuracy, and comparison against an automated analyzer is identified as the primary target for future clinical validation work.

The literature reviewed establishes that the proposed system is technically grounded, clinically justified, and

appropriately scoped within existing regulatory frameworks.
These foundations directly inform the design decisions
presented in Chapter 3.

CHAPTER 3 RESEARCH DESIGN AND METHODOLOGY

Description of the Proposed Study

This chapter presents the research design, system architecture, and evaluation framework for the development of an offline, edge-computing smartphone-based Clinical Decision Support System (CDSS) intended for automated urinalysis screening. The study adopts a developmental research design, a methodology specifically chosen to support the systematic design, development, and evaluation of a technological tool. This design is characterized by an iterative process that bridges software engineering with empirical validation, ensuring that the resulting system is both technically robust and contextually relevant. To guarantee rigorous algorithmic calibration prior to clinical testing, this methodology is implemented across two sequential phases.

The first phase consists of a controlled development and calibration stage, during which the system's illumination-adaptive image processing pipeline and probabilistic reasoning framework are constructed, calibrated, and internally evaluated using standardized synthetic urine control solutions and human urine specimens

collected from volunteers, both processed under predefined lighting conditions. This controlled environment enables systematic algorithm refinement while strictly minimizing uncontrolled biological variability. Subsequently, the second phase involves a structured validation process conducted in collaboration with a licensed healthcare facility. In this phase, the system's classification and risk stratification outputs are compared against the manual visual interpretations of licensed medical technologists using human urine specimens processed as part of a routine clinical workflow. This validation stage is designed to assess inter-method agreement under real-world ambient conditions rather than to establish population-level diagnostic performance.

The system is implemented under a strictly offline edge-computing architecture. All computational components, including image acquisition, illumination-adaptive preprocessing, region of interest (ROI) segmentation, HSV-based color feature extraction, k-Nearest Neighbors (k-NN) visual classification, structured symptom data integration, and Bayesian posterior probability estimation, are executed locally on the smartphone's onboard processor. No image data, clinical inputs, or probabilistic computations are

transmitted to external servers at any stage of operation. This architectural decision is intentional and aligned with the deployment context of Rural Health Units (RHUs), where reliable internet connectivity may not be consistently available.

All outputs generated by the system are explicitly framed as preliminary risk stratification for two predefined screening categories: urinary tract infection (UTI) and selected renal risk indicators. The system does not prescribe medication, does not provide definitive medical diagnoses, and is not evaluated as a substitute for laboratory analyzers, physician assessment, or confirmatory diagnostic procedures. The methodology remains strictly bounded by the scope, delimitations, and terminology established in the first chapter of this study.

Methods and Proposed Enhancements

The methodology of this study is architecturally grounded in a Late Fusion framework that integrates two functionally distinct computational modules: an illumination-adaptive computer vision pipeline and a probabilistic clinical reasoning engine. This hybrid structure directly addresses the two principal limitations identified in Chapter 1: (1) the sensitivity of smartphone-based colorimetric analysis to ambient lighting variability, and (2) the diagnostic limitations of unimodal image classification systems that interpret reagent pad color changes without incorporating symptom-informed clinical context.

The computer vision module is responsible for image acquisition, illumination-adaptive preprocessing, reagent pad segmentation, HSV-based color feature extraction, and k-Nearest Neighbors (k-NN) visual classification. The probabilistic reasoning module receives the k-NN classification outputs together with structured patient symptom inputs and applies Bayesian inference to compute posterior disease risk probabilities.

A Late Fusion strategy is adopted, wherein modality-specific outputs are independently computed before probabilistic integration. This design preserves the interpretability of each computational layer, maintains computational efficiency suitable for on-device inference, and avoids the instability that may arise from early fusion of heterogeneous feature types – continuous colorimetric vectors and categorical symptom variables – within a single feature space.

To achieve a comprehensive 10-parameter urinalysis profile (Leukocytes, Nitrite, Urobilinogen, Protein, pH, Blood, Specific Gravity, Ketone, Bilirubin, and Glucose), the system utilizes a Multi-Class Feature Mapping approach. The k-NN classifier is trained on 5,480 labeled HSV feature vectors derived from 175 volunteer urine specimens, each imaged under three controlled lighting conditions. This dataset provides sufficient colorimetric coverage across the four target analytes and their semi-quantitative grade levels to enable reliable nearest-neighbor classification within the predefined operational range.

The system utilizes a Flattened k-NN Search Architecture. By optimizing the feature vectors into 3D HSV coordinates, the application performs an exhaustive Brute

Force nearest-neighbor search across the 5,480-point reference dataset in under 200ms on a device with 4GB of RAM. This eliminates the need for complex tree-based partitioning while maintaining real-time performance for a full 10-parameter scan.

The succeeding subsections describe each methodological component according to its position within the operational pipeline – from image capture and illumination correction, through feature extraction and visual classification, to probabilistic disease risk estimation, validation procedures, and deployment architecture.

System Architecture Overview

The proposed system follows a sequential five-stage pipeline architecture implemented entirely on-device.

Stage 1 – Image Acquisition:

The smartphone camera captures an image of the reacted reagent dipstick under ambient indoor lighting conditions within the predefined operational range (2700K-5500K). The app supports both live camera capture via CameraController and gallery selection via ImagePicker.

Stage 2 - Illumination-Adaptive Preprocessing (AWB):

An Adaptive White-Balance (AWB) algorithm performs reference-based color normalization using the unreacted white plastic substrate of the dipstick as a calibration anchor. This step reduces color temperature-induced bias prior to feature extraction, while acknowledging that residual lighting variability may persist.

Stage 3 - Region of Interest (ROI) Segmentation:

Automated segmentation isolates the ten chemically treated reagent pads from the background using flood-fill connected component analysis (`_floodDarkComponent`), with pad positions placed geometrically via `_findBestPadGrid` and `_refinePadY` relative to the detected marker. Fully automatic – no user interaction required. This step ensures that subsequent color analysis is restricted to the relevant reaction regions and not influenced by extraneous background pixels.

Stage 4 - HSV Feature Extraction and k-NN Classification:

For each isolated pad, mean HSV feature vectors are computed and submitted to the k-NN classifier. The classifier returns both a class label and a neighbor-

proportion probability, representing a soft confidence estimate of the pad's semi-quantitative reaction level.

Stage 5 - Bayesian Late Fusion and Risk Stratification:

The Bayesian reasoning module integrates the per-pad visual probabilities with structured clinical symptom inputs to compute posterior probabilities for the predefined screening categories: urinary tract infection (UTI) and selected renal risk indicators. These posterior probabilities are subsequently mapped to discrete risk stratification levels presented to the user.

All five stages are executed sequentially and locally on the device, without reliance on external servers, cloud-based processing, or network connectivity. This pipeline represents the complete operational workflow of the CDSS, from image capture to preliminary risk output.

Minimum Viable Hardware (MVH) Specifications

To standardize the operational definition of 'budget-tier edge computing,' the application is constrained to run on devices meeting the following Minimum Viable Hardware (MVH) specifications: an Android operating system (Version 8.0 Oreo or higher), a minimum of 4GB Random Access Memory

(RAM) to safely execute the k-NN nearest-neighbor distance calculations across the reference dataset without Out-Of-Memory (OOM) errors, and a rear-facing camera sensor of at least 8 Megapixels with autofocus capabilities. Devices meeting these baseline specifications represent the ubiquitous, low-cost smartphones readily accessible to primary healthcare workers in Philippine RHUs.

Data Acquisition and Training Dataset Construction

Algorithm development, k-NN classifier training, and Adaptive White-Balance (AWB) calibration are conducted primarily using human urine specimens collected from healthy volunteers and processed by licensed medical technologists at a partner Rural Health Unit (RHU). Preliminary colorimetric separability was established using commercially prepared synthetic control solutions prior to human sample collection.

Human Urine Specimens – Training Dataset

A total of 175 volunteer urine specimens were collected from healthy volunteers and forwarded to the partner RHU for processing under standard laboratory protocols. Specimen collection, handling, processing, and disposal were conducted exclusively by licensed healthcare personnel

at the partner facility. The researchers did not directly collect, store, transport, or manipulate biological specimens at any stage.

Each processed dipstick was imaged under the three predefined lighting conditions, 2700K (Warm White), 4000K (Neutral White), and 5500K (Cool Daylight), yielding 3 images per sample and a total of 5,480 labeled HSV feature vectors for classifier training and cross-validation. Each image was labeled according to the semi-quantitative reaction grade independently determined by the licensed medical technologist at the partner facility across the four target analytes: Glucose, Protein, Leukocyte Esterase, and Nitrite.

No synthetic augmentation or oversampling was applied to the training dataset. The 5,480 labeled feature vectors represent the complete unmodified reference space upon which the k-NN classifier operates.

For each sample and lighting condition, images were captured at a standardized distance, orientation, and approximate framing to minimize geometric variability. Each strip reading follows the manufacturer's dip-and-read protocol, immersing the strip for two seconds and reading

results at 60 seconds, to minimize kinetic variability across samples [8][69].

These labeled samples collectively form the structured reference dataset used for:

Estimating AWB gain factors under varying lighting conditions

Constructing HSV feature distributions for each target analyte

Training and cross-validating the k-NN classifier

This dataset establishes the foundational colorimetric reference space upon which the validation phase described in Section 3.2 is performed.

The system utilizes a Flattened k-NN Search Architecture to process the 10-parameter reagent strip. By optimizing the reference feature vectors into 3D HSV coordinates, the application performs an exhaustive Brute Force nearest-neighbor search across the 5,480-point reference dataset. This architectural choice maintains a computational latency

of under 200ms, ensuring real-time diagnostic feedback without the need for complex tree-based partitioning such as k-d trees, which can be structurally unstable on low-memory mobile environments.

Rationale for Parameter Delimitation

The software architecture differentiates between Broad-Spectrum Urinalysis and Targeted Clinical Risk Profiling. While the Perception Layer captures and displays all 10 chemical metrics to the user for a complete laboratory record, only the four primary 'Sentinel Analytes' (Glucose, Protein, Leukocytes, and Nitrites) are fed into the Bayesian Late Fusion reasoning engine. This is a deliberate clinical design choice: the Sentinel Analytes provide the highest likelihood ratios for the study's primary objectives (UTI and NCD screening), whereas the remaining six parameters serve as supplemental data for the clinician's manual review, ensuring a balance between automated risk estimation and comprehensive diagnostic transparency.

While the physical reagent strip utilized in this study captures 10 distinct chemical parameters, the computer vision and the k-NN classifier processes all ten

analyte pads and all ten results are displayed to the user. Only the four Sentinel Analytes feed into the Bayesian engine: Glucose, Protein (Albumin), Leukocyte Esterase, and Nitrite. This methodological scoping is anchored on three foundational constraints of the study: Clinical Relevance, Mathematical Integrity, and Hardware Limitations.

1. Clinical Alignment with Primary Care Guidelines

The target deployment environment for this Clinical Decision Support System (CDSS) is resource-constrained Rural Health Units (RHUs). According to the Philippine Package of Essential NCD Interventions (PhilPEN) guidelines [3, 4], the primary mandate of RHUs is the early triage of Non-Communicable Diseases (NCDs) and common bacterial infections. Glucose and Protein serve as the definitive screening markers for Diabetes and Chronic Kidney Disease (CKD), while Leukocytes and Nitrite are the primary indicators for Urinary Tract Infections (UTIs). The remaining six parameters on the standard dipstick (e.g., Bilirubin, Urobilinogen, Ketones) indicate complex hepatic or metabolic pathologies that exceed the treatment capacity of primary RHUs and necessitate hospital referral. Thus, restricting the algorithm to these four critical parameters

aligns the software precisely with its intended clinical triage utility.

2. Edge-Computing Constraints on Budget Hardware

Finally, the system is architected to operate entirely offline on standard, budget-tier Android smartphones to accommodate regions without reliable internet connectivity. The k-NN algorithm relies on real-time distance computations against the stored dataset. Expanding the feature space to accommodate the reference points required for 10-parameter classification would exponentially increase the computational load. This would likely result in Out-Of-Memory (OOM) errors, thermal throttling, or unacceptable diagnostic latency on low-RAM devices. The 4-parameter scope ensures that the CDSS executes instantaneously and reliably within the strict hardware constraints of edge computing.

Adaptive White-Balance (AWB) for Illumination-Adaptive Processing

A primary source of classification instability in smartphone-based colorimetric analysis is illumination-

induced color shift, wherein the same chemically reacted reagent pad produces different RGB values under varying ambient color temperatures. Although this phenomenon is often informally attributed to metamerism, the practical concern in smartphone imaging arises from changes in illuminant spectral composition and camera auto-adjustments. The smartphone's native automatic white-balance (AWB) algorithm is optimized for perceptual image aesthetics rather than quantitative colorimetric consistency, and therefore cannot be relied upon for controlled analytical normalization.

To mitigate this variability, the system implements a reference-based Adaptive White-Balance (AWB) algorithm grounded in a marker-patch gain correction. Rather than estimating scene illumination from global image statistics, which can be unreliable in non-neutral scenes, the algorithm uses the unreacted center patch of the black reference square printed on the dipstick of the dipstick as a physically stable illumination reference. This region does not undergo chemical reaction and maintains a near-neutral reflectance under standard conditions.

Let the mean RGB values of the detected white reference region be denoted as: $\bar{R}_{ref}, \bar{G}_{ref}, \bar{B}_{ref}$

Under ideal neutral illumination and camera response, this region would approximate a high-intensity neutral white. Deviations from neutrality are therefore interpreted as illumination-induced channel bias.

Per-channel gain factors are computed as:

$$G_R = \frac{R_{target}}{\bar{R}_{ref}}, \quad G_G = \frac{G_{target}}{\bar{G}_{ref}}, \quad G_B = \frac{B_{target}}{\bar{B}_{ref}}$$

where $R_{target}, G_{target}, B_{target}$ represent the desired neutral reference values. In practice, these are set to 255 for 8-bit RGB normalization.

Each pixel in the image is then adjusted multiplicatively:

$$R' = G_R \cdot R, \quad G' = G_G \cdot G, \quad B' = G_B \cdot B$$

This transformation produces a color-corrected image in which the white reference region approximates neutral white and all other regions—including reagent pads—are proportionally normalized. By performing this correction prior to segmentation and feature extraction, the system reduces systematic color bias introduced by ambient illumination.

It is important to emphasize that this AWB implementation provides illumination-adaptive normalization rather than absolute illumination invariance. Residual variability may persist due to factors such as specular reflections on wet pad surfaces, uneven illumination gradients across the strip, sensor noise, or lighting conditions outside the predefined operational range of 2700K-5500K.

Accordingly, illumination robustness is evaluated empirically rather than assumed. Stability is quantified by measuring the consistency of classification outputs across the three predefined lighting conditions during the calibration phase. This evaluation approach ensures that illumination adaptation is treated as a measurable performance characteristic rather than an absolute property of the system.

Region of Interest (ROI) Segmentation

Following illumination-adaptive preprocessing, the system performs automated Region of Interest (ROI) segmentation to isolate the pixel regions corresponding to each of the ten reagent pads from the surrounding

background. This step is essential because variations in camera distance, strip placement, orientation, and background texture may otherwise introduce irrelevant pixel data into the feature extraction process. Direct full-image color analysis would therefore reduce classification reliability by incorporating non-reactive regions.

ROI segmentation is implemented through a structured two-stage procedure.

Stage 1 - Dipstick Localization:

Adaptive thresholding is applied to the color-corrected image to generate a binary mask that differentiates the dipstick structure from the background. Unlike global thresholding, adaptive thresholding computes local intensity thresholds across image neighborhoods, allowing improved separation under uneven lighting conditions. This produces a preliminary binary segmentation highlighting the elongated strip geometry.

Subsequently, contour detection is performed on the thresholded mask to identify candidate geometric boundaries. The contour corresponding to the dipstick strip is selected based on geometric constraints such as area, aspect ratio, and elongated rectangular structure. This

step establishes the spatial boundary of the strip within the image frame.

Stage 2 - Pad Region Extraction:

Once the strip boundary is identified, the image is optionally aligned or normalized relative to the detected orientation to reduce rotational variability. The ten reagent pads are then extracted as fixed relative sub-regions along the longitudinal axis of the strip based on the known physical layout of the standard 10-parameter dipstick configuration.

Each pad region is defined using proportional positional offsets within the detected strip boundary rather than absolute pixel coordinates, ensuring robustness to moderate variations in scale and framing. The segmented pad regions are then individually forwarded to the HSV feature extraction stage for colorimetric analysis.

This ROI segmentation strategy constrains subsequent feature extraction strictly to chemically reactive areas of interest, thereby improving classification consistency and minimizing background-induced noise.

HSV Feature Vector Construction

Color features are extracted from each segmented reagent pad in the HSV (Hue, Saturation, Value) color space rather than the standard RGB color space. This representation is selected because the Hue and Saturation channels encode the chromatic properties of the pad color change—which correspond to the chemical reaction of interest—while the Value channel encodes brightness intensity, which is more susceptible to residual illumination variation. By operating primarily on Hue and Saturation, the classifier focuses on the chemical signal while reducing sensitivity to brightness-related noise that the AWB correction may not have fully eliminated.

For each segmented pad region, the mean Hue, mean Saturation, and mean Value are computed across all pixels within the region, yielding a three-dimensional HSV feature vector $\mathbf{x} = [H, S, V]$ per pad. These feature vectors serve as the input to the k-NN classifier in the subsequent stage.

Dual-Stage Classifier: k-Nearest Neighbors (k-NN) and Bayesian Reasoning

The classification architecture is structured as a deliberate dual-stage design that separates the perceptual

task of colorimetric pattern recognition from the inferential task of disease-level probabilistic reasoning. This separation is methodologically necessary because k-Nearest Neighbors (k-NN), operating in HSV feature space, is well suited to identifying geometric proximity between an observed pad color and calibrated reference color clusters, but it does not inherently account for the asymmetric diagnostic importance of individual dipstick parameters nor allow principled integration of heterogeneous symptom variables. The Bayesian model in Stage 2 explicitly addresses these limitations.

Stage 1 – k-Nearest Neighbors (k-NN) Visual Classification

The k-NN algorithm functions as the system's Perception Layer, operating on HSV feature vectors extracted from each segmented reagent pad.

Because the HSV color space is cylindrical, utilizing a standard Euclidean distance metric would mathematically misrepresent the Hue channel (where 0° and 360° both represent red). To resolve this geometric inconsistency, the k-NN algorithm employs a Circular Distance Metric specifically for the Hue dimension. The angular distance between any two hue vectors (H_1 and H_2) is calculated

using the shortest path around the cylinder: $d(H_1, H_2) = \min(|H_1 - H_2|, 360 - |H_1 - H_2|)$. This custom distance metric prevents the classifier from incorrectly differentiating between visually identical chromatic ends, ensuring accurate colorimetric matching against the reference dataset

The k nearest neighbors are identified based on smallest distance values. The predicted class for the pad is determined through majority voting among these neighbors.

Rather than producing a hard classification only, the model computes a soft visual probability estimate:

$$P_{\text{visual}}(C_j \mid x) = (\text{Number of neighbors belonging to class } C_j) / k$$

This neighbor-proportion probability represents the visual confidence associated with each semi-quantitative reaction level. These per-pad visual probabilities are then forwarded to the Bayesian Late Fusion module.

The value of k is selected through cross-validation on the training dataset to balance sensitivity to subtle color

differences against robustness to noise and lighting residuals.

Stage 2 – Bayesian Late Fusion for Probabilistic Disease

Risk Estimation

The Bayesian model constitutes the system's Reasoning Layer. While k-NN operates independently per pad in geometric feature space, the Bayesian model operates at the disease level, integrating:

- Multi-parameter visual outputs
- Symptom inputs
- Prior disease prevalence

The posterior probability of disease condition H given evidence E is computed using Bayes' Theorem:

$$P(H \mid E) = [P(E \mid H) \times P(H)] / P(E)$$

Prior Probability – P(H): The prior probability represents the baseline prevalence of the condition within the target screening population before observing test results.

For:

- UTI – prevalence estimates from primary care populations

- Basic renal impairment indicators – epidemiological estimates relevant to the demographic context

These priors are fixed during system initialization and remain constant during individual screenings. They serve as baseline risk anchors rather than dynamic predictors.

Likelihood – $P(E|H)$: The likelihood represents the probability of observing the collected evidence given the presence (or absence) of the disease.

Evidence variables include:

- Visual pad results (derived from k-NN probabilities)
- Symptom inputs (e.g., dysuria, urgency, fever)

Each dipstick parameter is parameterized using clinically reported sensitivity and specificity values. For example:

- Nitrite – high specificity for UTI, contributing strong positive likelihood ratio
- Leukocyte esterase – higher sensitivity but lower specificity, contributing complementary diagnostic weight

Symptom variables are incorporated as binary indicators with conditional probabilities grounded in documented clinical associations.

Structured Likelihood Computation with Feature Grouping

The combined likelihood is computed as the product of individual conditional probabilities across independent evidence variables:

$$P(E \mid H) = \prod P(e_i \mid H)$$

However, evidence variables with known clinical correlation are not treated as independent. Specifically, variables such as Nitrite and Leukocyte Esterase in the context of UTI, and overlapping symptoms such as Dysuria and Urgency, are managed via Feature Grouping. Correlated evidence is merged into joint probability nodes prior to likelihood multiplication. This prevents the double-counting of correlated evidence, mitigating the risk of artificially inflated posterior probabilities and ensuring a more conservative, clinically reliable risk estimation.

Posterior Risk Stratification: The computed posterior probability $P(H \mid E)$ is mapped into discrete screening categories:

- Low Risk
- Moderate Risk
- High Risk

These outputs are explicitly framed as preliminary risk stratification, not diagnosis.

All Bayesian computations are executed locally on-device. No cloud inference or external communication occurs at any stage.

Structured Clinical Evidence and Symptom Vectorization

To operationalize the Bayesian inferential stage, the system employs a structured clinical checklist that transforms qualitative patient-reported symptoms into a quantitative Symptom Vector (E). This methodological choice deliberately replaces Natural Language Processing (NLP) to eliminate linguistic ambiguity and ensure computational efficiency within an offline, resource-constrained mobile environment. By utilizing a binary input format, the "Reasoning Layer" achieves a deterministic mapping of clinical evidence, which is essential for maintaining the integrity of the posterior probability update.

A. Structured Clinical Symptom Checklist

The following symptoms are presented to the user as a binary (Yes/No) selection interface. These variables represent the clinical evidence used to update the posterior probability of the screening result:

Category 1: Lower Urinary Tract Symptoms (UTI Focus)

- Dysuria: Persistent pain, burning, or stinging sensation during urination.
- Urinary Frequency/Urgency: A sudden, strong, or frequent need to urinate.
- Suprapubic Pain: Pain or pressure felt in the lower abdomen or pelvic area.
- Visible Hematuria: Urine that appears visibly red, pink, or significantly cloudy.

Category 2: Renal and Systemic "Red Flags"

- Flank Pain: Sharp or dull pain located in the side of the back, below the ribcage.
- Fever/Chills: Body temperature exceeding 38°C, indicating a systemic immune response.
- Peripheral Edema: New or unusual swelling in the ankles, feet, or periorbital (eye) area.
- Nausea/Vomiting: Persistent nausea often associated with acute renal stress or systemic infection.

B. Bayesian Likelihood Ratios and Diagnostic Weighting

The "asymmetric diagnostic importance" is mathematically integrated through the application of Likelihood Ratios (LR). These ratios represent the strength of the clinical evidence in favor of a positive screening result. In the Bayesian update rule, a symptom with an $LR > 1.0$ increases the post-test probability, whereas the absence of symptoms ($LR < 1.0$) effectively moderates the risk, accounting for potential asymptomatic bacteriuria or false-positive chemical interference.

**TABLE IV DIAGNOSTIC WEIGHTS (LIKELIHOOD RATIOS) FOR
CLINICAL EVIDENCE**

Clinical Domain	Symptom Variable	Likelihood Ratio (LR)	Bayesian Interpretant
Lower UTI	Dysuria	3.0	High correlation with infection
	Frequency / Urgency	1.8	Moderate supportive evidence
	Suprapubic Pain	1.5	Low-moderate supportive evidence
	Visible Hematuria	2.0	Strong correlate for Blood pad
Upper UTI	Flank Pain	5.0	High-weight critical indicator

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	Fever / Chills	3.5	Indicator of systemic involvement
	Peripheral Edema	4.0	High weight for renal impairment
	Nausea / Vomiting	2.2	Supportive systemic evidence

To ensure the clinical validity of the Bayesian reasoning engine, the symptom Likelihood Ratios (LRs) outlined in Table IV were not arbitrarily assigned, but rather derived from established urological meta-analyses and systematic reviews. The probabilistic weights for lower urinary tract symptoms—specifically Dysuria (LR+ 1.5 to 3.0), Urinary Frequency/Urgency (LR+ 1.8), and Visible Hematuria (LR+ 2.0)—are anchored on the foundational diagnostic frameworks established by Bent et al. (2002) in their systematic review published in *JAMA* ("*Does this woman have an acute uncomplicated urinary tract infection?*") [71].

Symptoms are categorized into Lower Urinary Tract Symptoms (LUTS), indicating localized bladder irritation, and Upper Urinary Tract / Systemic Red Flags, which represent potential pyelonephritis or renal involvement. This distinction is reflected in the Bayesian Likelihood Ratios, where Upper UTI symptoms (e.g., Flank Pain,

\$LR=5.0\$) are weighted more heavily to reflect increased clinical urgency.

The higher Likelihood Ratios assigned to systemic and upper-tract red flags, such as Flank Pain (LR+ 5.0) and Fever/Chills (LR+ 3.5), reflect their established clinical significance as indicators of pyelonephritis and systemic involvement, consistent with published triage prioritization guidelines for upper urinary tract conditions. This weighting ensures that the Bayesian reasoning engine properly stratifies patients based on clinically validated pre-test probabilities.

Risk Stratification Threshold Design

The transition from continuous posterior probabilities to discrete, actionable classifications is governed by a tripartite threshold model. These thresholds serve as the final decision logic of the reasoning layer, categorizing the fused multimodal output into Low, Moderate, or High Risk profiles. This categorization ensures that the screening results are interpretable for non-clinical users in Rural Health Units (RHUs) while maintaining the statistical rigor required for clinical decision support.

A. Categorical Risk Definitions

The threshold boundaries are defined based on the accumulation of visual and clinical evidence (E), and are consistent with published UTI diagnostic frameworks that employ analogous probability boundaries to stratify patients using combined clinical and dipstick-based evidence [72]:

Low Risk ($P < 0.30$): This profile is assigned when both the colorimetric analysis (k-NN) returns results within the physiological baseline and the clinical checklist indicates an absence of symptomatic distress. This represents a high negative predictive value for the targeted conditions.

Moderate Risk ($0.30 < P < 0.70$): This classification occurs under conditions of "Clinical Divergence" – such as when a reagent pad indicates a trace positive but the patient is asymptomatic, or when a patient reports mild symptoms despite inconclusive visual data. This category acts as a monitoring buffer, suggesting a follow-up test.

High Risk ($P > 0.70$): This profile is triggered by strong positive correlation between reagent pad intensity and high-weight clinical symptoms (e.g., Flank Pain or Fever). Due to the high Likelihood Ratios (LR) assigned to

"Red Flag" symptoms, the presence of systemic indicators will often escalate the profile to High Risk even in the presence of low-concentration visual signals.

Currently, the stratification thresholds of 0.30 and 0.70 serve as heuristic baselines calibrated to enforce a 'safety-first' diagnostic weighting. However, to ensure these decision boundaries are statistically sound, the exact numerical cut-offs will be empirically optimized during the Phase 2 clinical validation, wherein ROC curve analysis and Youden's J Statistic will be applied to the clinical dataset to mathematically determine the optimal threshold points, maximizing diagnostic accuracy by balancing Sensitivity and Specificity across varying patient presentations.

B. Threshold Calibration and Sensitivity

The boundaries are optimized to prioritize sensitivity in identifying potential renal impairment and upper UTIs. By setting the High-Risk threshold at 0.70, the system ensures that any combination of a positive chemical analyte (Nitrite/Leukocytes) and a primary systemic symptom (Flank Pain) mathematically forces the posterior probability beyond the critical boundary, consistent with the system's

objective as a screening tool designed to prevent the progression of undetected infections into severe renal complications.

Components and Design

System Architecture

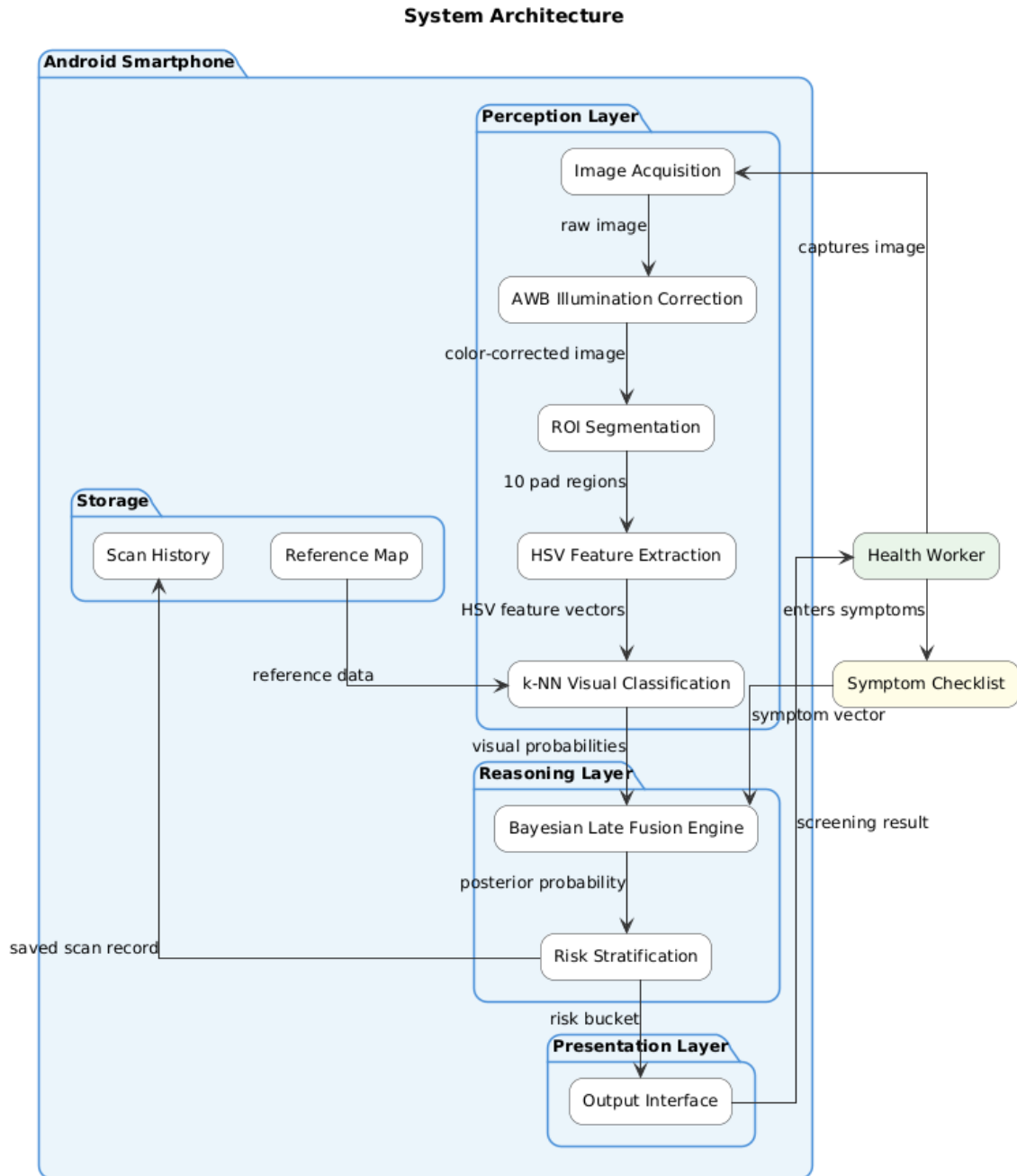


Fig. 1. System Architecture

Figure 1 presents the system architecture of Uritect as a two-layer block diagram operating entirely on an Android smartphone. The Perception Layer processes the captured dipstick image through a sequential pipeline: AWB illumination correction using a black reference marker patch, pad grid localization relative to the detected marker, per-pad HSV feature extraction using circular mean hue, and k-NN classification against a bundled reference map of 5,480 samples at k=31. The Reasoning Layer receives screening probabilities from four sentinel analytes (GLU, LEU, PRO, NIT) and combines them with a user-supplied symptom checklist through Bayesian log-odds fusion, producing a posterior probability and a risk bucket of Low (below 0.30), Moderate (0.30 to 0.70), or High (above 0.70). All 10 analytes are classified and presented to the user, but only those four feed the posterior computation. The system requires no internet connection, no external server, and no user alignment guidance – all inference runs on-device from a single captured image.

Software Architecture

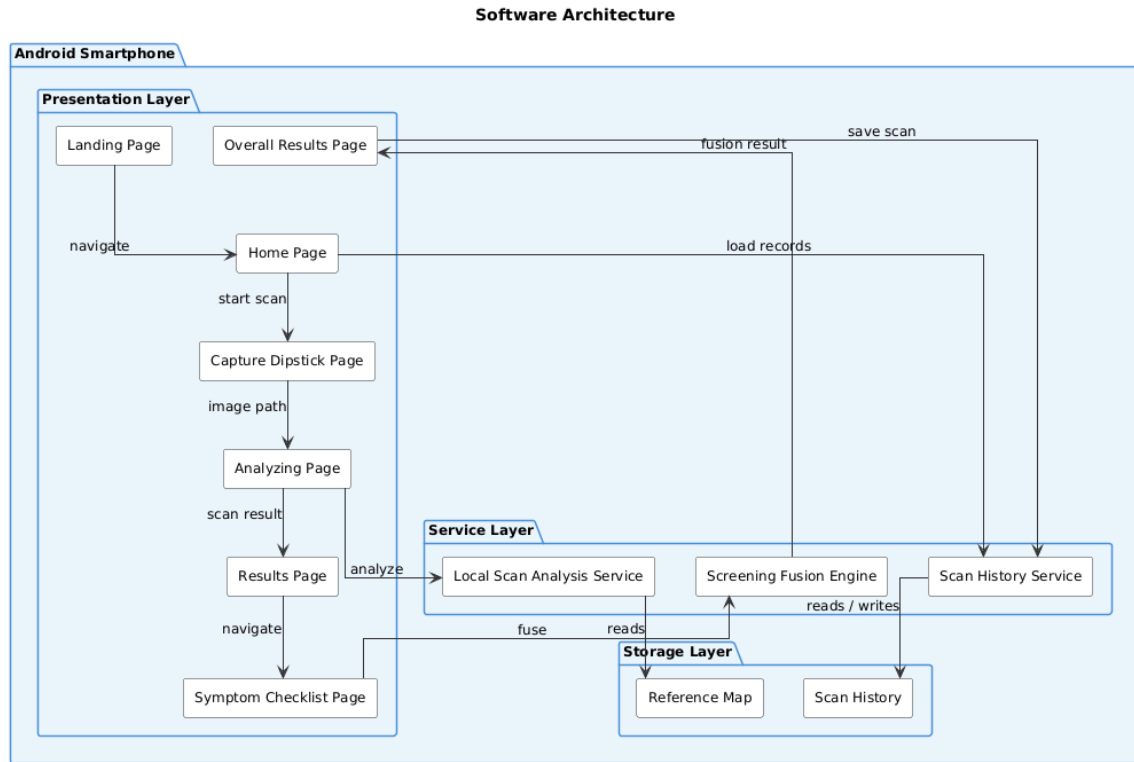


Fig. 2. Software Architecture

Figure 2. illustrates the layered software architecture of the Uritect Flutter application running on an Android smartphone. The Presentation Layer comprises seven pages: `LandingPage`, `HomePage`, `CaptureDipstickPage`, `AnalyzingPage`, `ResultsPage`, `SymptomChecklistPage`, and `OverallResultsPage`, each responsible only for UI rendering and navigation. The Service Layer contains three classes that hold all business logic: `LocalScanAnalysisService` runs the complete computer vision pipeline inside a Dart `Isolate`, `ScreeningFusionEngine` performs Bayesian log-odds

fusion, and ScanHistoryService reads and writes the persistent scan record to the app documents directory. The Storage Layer consists of two JSON files. knn_reference_map.json is bundled as a read-only asset at build time and is never modified on device, while scan_history/history.json is an append-only flat array written via path_provider after each completed scan.

Data Dictionary

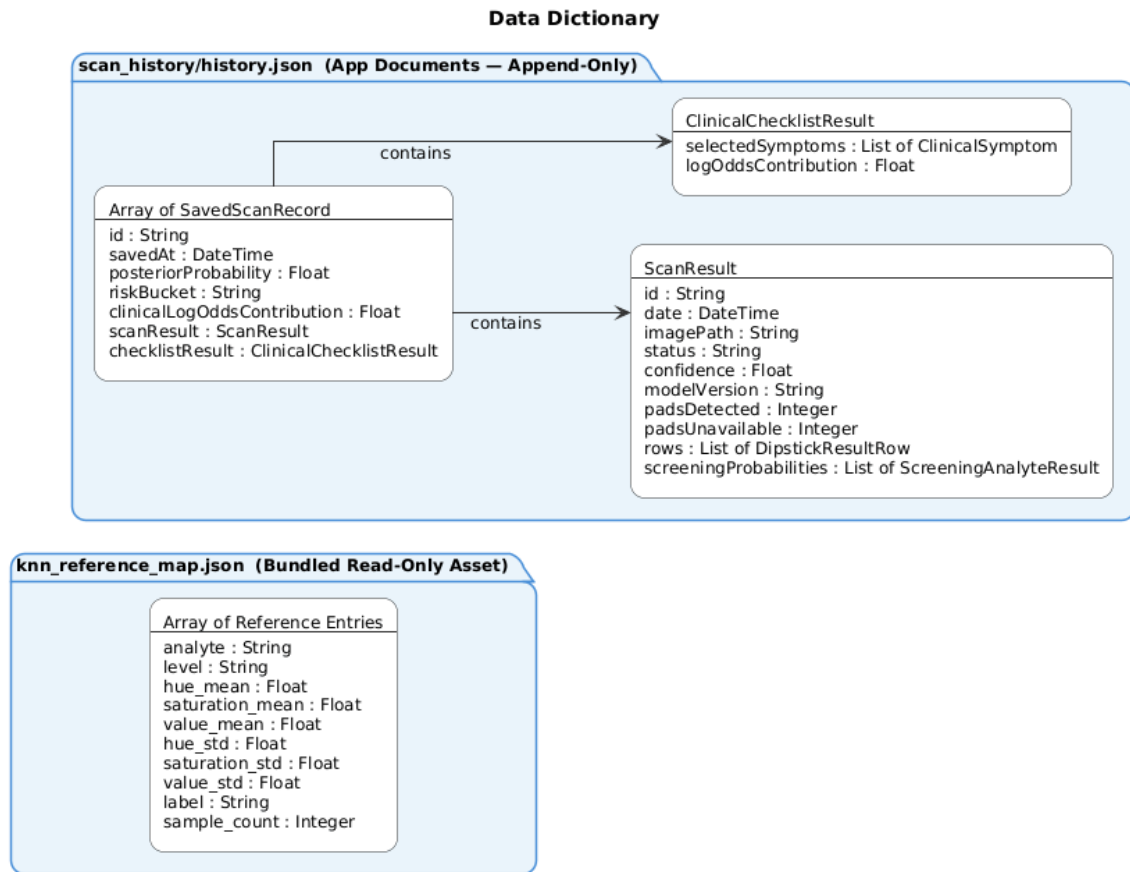


Fig. 3. Data Dictionary Diagram

Figure 3. presents the data dictionary for the two JSON document structures that constitute the entire storage layer of Uritect. The `knn_reference_map.json` file is a bundled read-only asset containing an array of reference entries, each specifying analyte name, concentration level label, mean and standard deviation for hue, saturation, and value, and a sample count; the file carries version `trained_v4_hsv`, records `k=31`, and holds 5,480 total

samples. The `scan_history/history.json` file is an append-only array of `SavedScanRecord` objects written to the application documents directory after each scan; each record embeds a full `ScanResult` containing all ten `DipstickResultRow` entries and the `screeningProbabilities` map for the four sentinel analytes, alongside a `ClinicalChecklistResult` holding the symptom selection state and log-odds contribution. A data dictionary was chosen over an entity-relationship diagram because the real data layer is document-based and does not involve relational tables, foreign keys, or any form of database engine.

Procedural Design

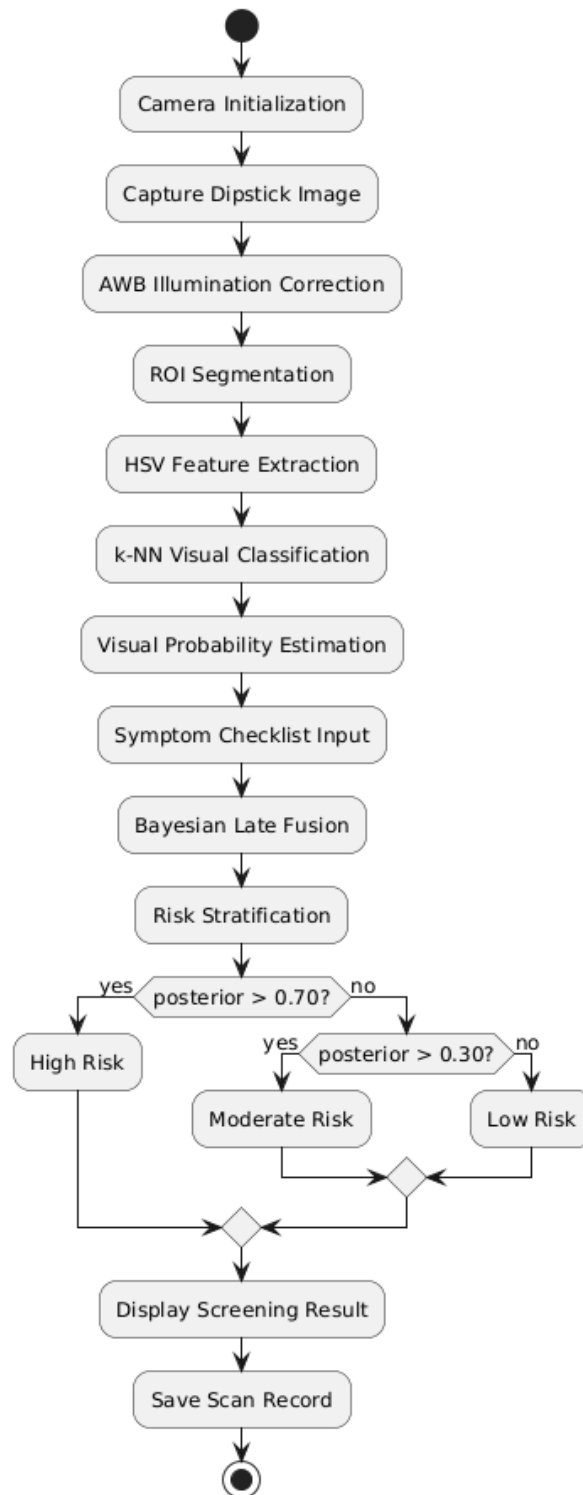


Fig. 4. Procedural Diagram

Figure 4. shows the procedural design of Uritect as a linear flowchart from camera initialization through to scan record persistence. After the user captures or selects an image, the system loads the reference map, runs flood-fill marker detection, applies AWB gain correction from the marker center patch, locates the 10-pad grid using a brute-force search across x-offset and y-offset space, extracts mean HSV features per pad with circular hue averaging, and runs k-NN classification to produce a level and screening probability for each analyte. The user views the analyte results and proceeds to fill the symptom checklist, after which Bayesian fusion computes a posterior probability and applies the risk thresholds: posterior above 0.70 yields High Risk, 0.30 to 0.70 yields Moderate Risk, and below 0.35 yields Low Risk. The scan record is saved and the result is displayed. The entire CV pipeline after image submission is automatic with no user alignment step required, as the 4-orientation brute-force rotation handling resolves dipstick orientation without user intervention.

Object-Oriented Design

Camera Layer

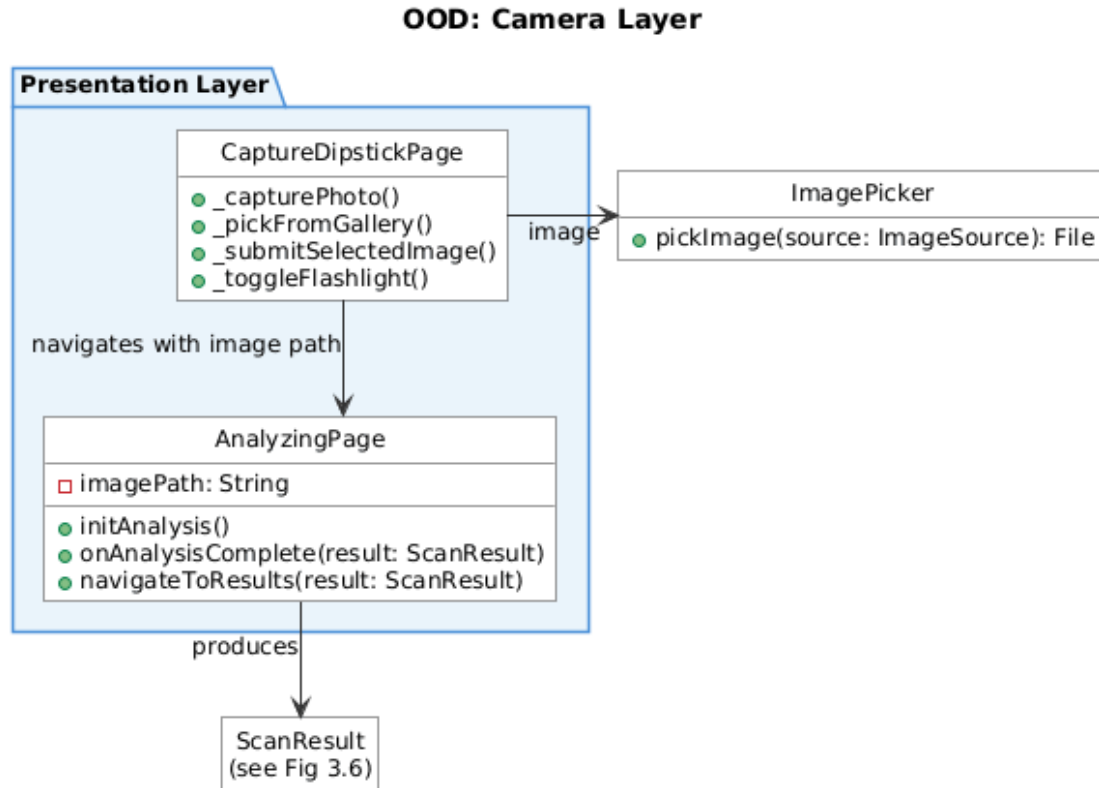


Fig. 5. OOD of Camera Layer

Figure 5 presents the object-oriented design of the camera and image acquisition layer. **CaptureDipstickPage** exposes three core interactions: `_capturePhoto()` which uses the device **CameraController** at high resolution, `_pickFromGallery()` which invokes **ImagePicker** for retrospective image selection, and `_submitSelectedImage()` which navigates to **AnalyzingPage** with the selected image path; a fourth method, `_toggleFlashlight()`, controls torch mode on the **CameraController**. There is no overlay alignment

UI – the user captures the dipstick freely within the camera preview. AnalyzingPage receives the image path, immediately invokes LocalScanAnalysisService, and navigates to ResultsPage upon completion, passing the resulting ScanResult object whose full structure is detailed in Figure 6.

Visual And Classification Layer

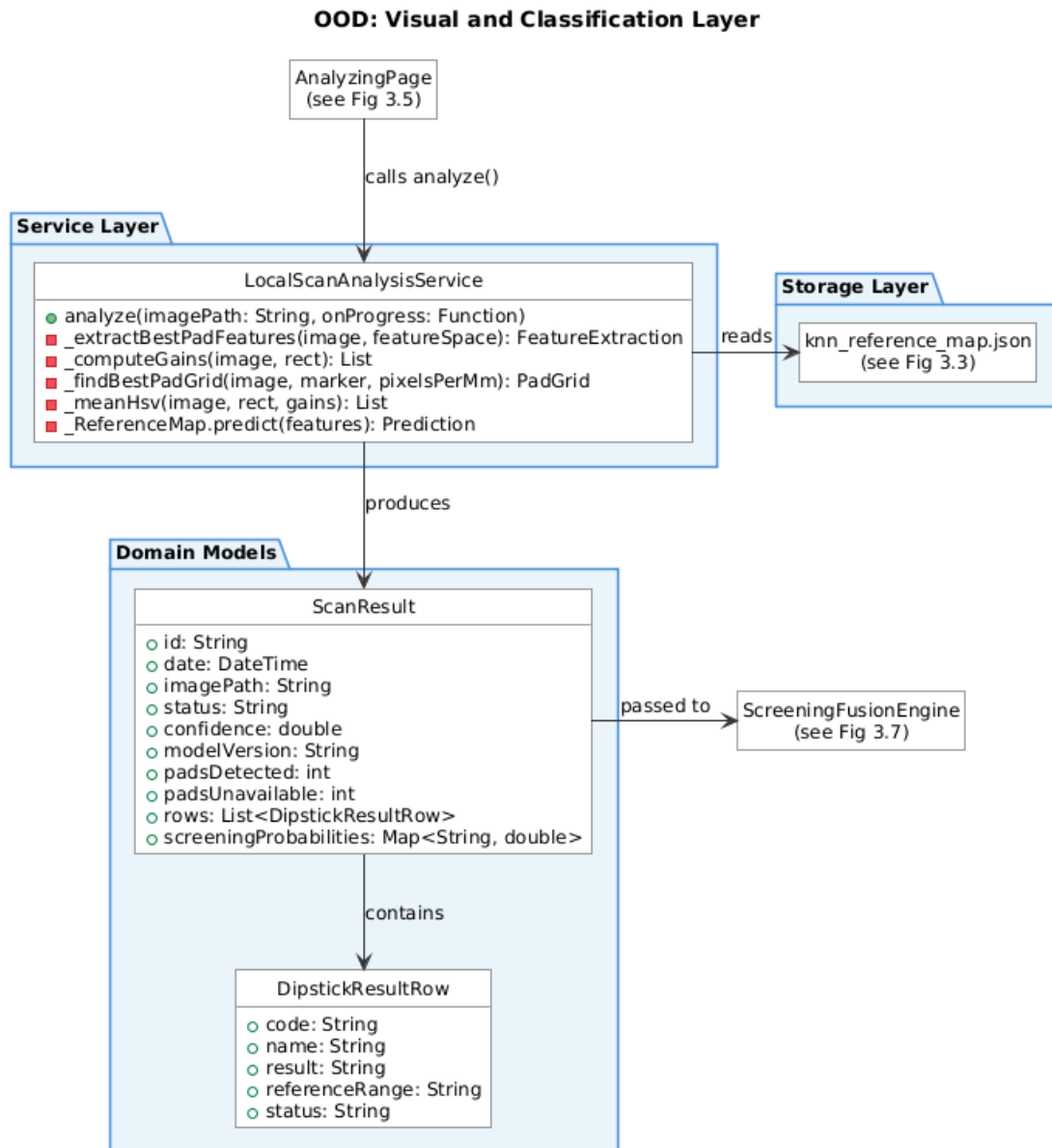


Fig. 6. OOD of Visual and Classification Layer

Figure 6 details the object-oriented design of the visual analysis and k-NN classification layer. `LocalScanAnalysisService` is the core class, executing the full CV pipeline through `analyze()`, which dispatches work

to a Dart Isolate; internally, `_markerCandidates()` and `_floodDarkComponent()` perform flood-fill connected component analysis to locate the black reference square, `_extractBestPadFeatures()` tries four image orientations (0° , 90° , 180° , 270°) and selects the best by quality score, `_computeGains()` samples the marker center patch to compute per-channel AWB gain ratios, `_findBestPadGrid()` and `_refinePadY()` locate and fine-tune the 10 pad positions relative to the marker, and `_ReferenceMap.predict()` performs Euclidean k-NN with inverse-distance weighted voting against the reference entries loaded from `knn_reference_map.json`. The service produces a `ScanResult` containing all ten `DipstickResultRow` entries and a `screeningProbabilities` map restricted to GLU, LEU, PRO, and NIT, which is then passed to `ScreeningFusionEngine` as described in Figure 7.

Reasoning Layer

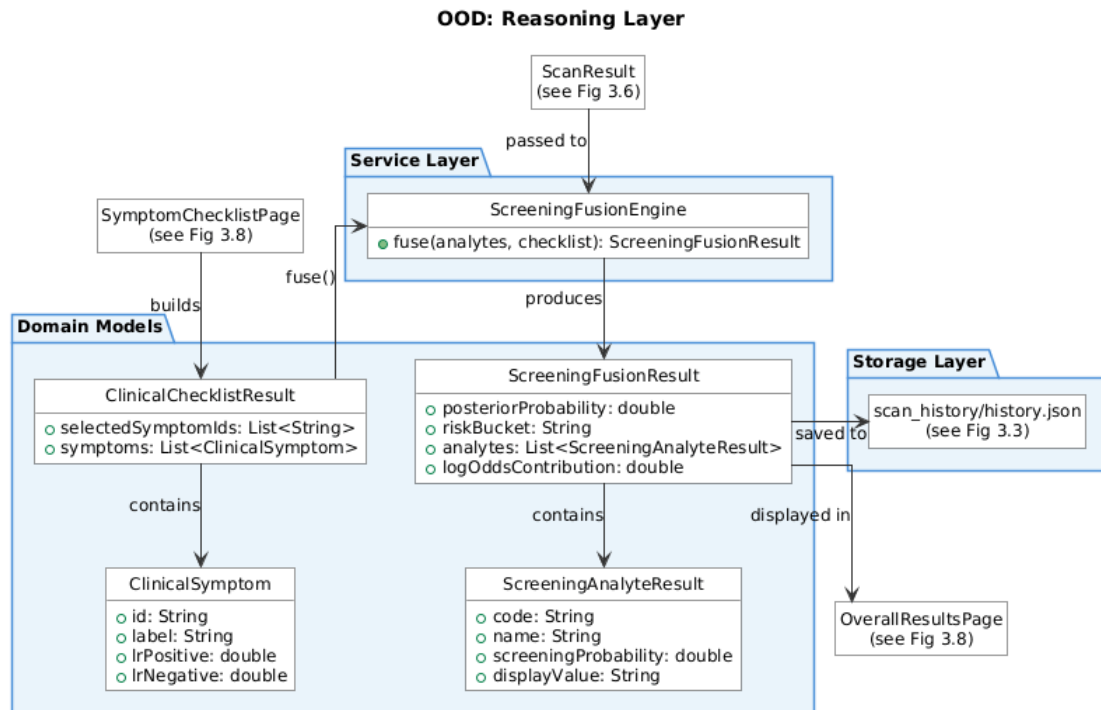


Fig. 7. OOD of Reasoning Layer

Figure 7 presents the object-oriented design of the Bayesian reasoning and persistence layer.

ScreeningFusionEngine receives a list of four ScreeningAnalyteResult objects and a ClinicalChecklistResult, then computes a posterior probability by converting the mean pad abnormal probability to log-odds, adding the log-sum of per-symptom likelihood ratios from the ClinicalChecklistResult, and transforming the result back to probability space.

ClinicalChecklistResult holds the user's symptom selections as a map from symptom ID to boolean and computes the log-odds contribution via computeLogOddsContribution(),

iterating over the eight hardcoded ClinicalSymptom entries, each carrying a label, LR+, and LR-. The fusion output is a ScreeningFusionResult containing the posterior probability, risk bucket (Low, Moderate, or High), the four ScreeningAnalyteResult entries, and the clinical log-odds contribution, which SymptomChecklistPage uses to call ScanHistoryService before navigating to OverallResultsPage.

Use Case Diagram

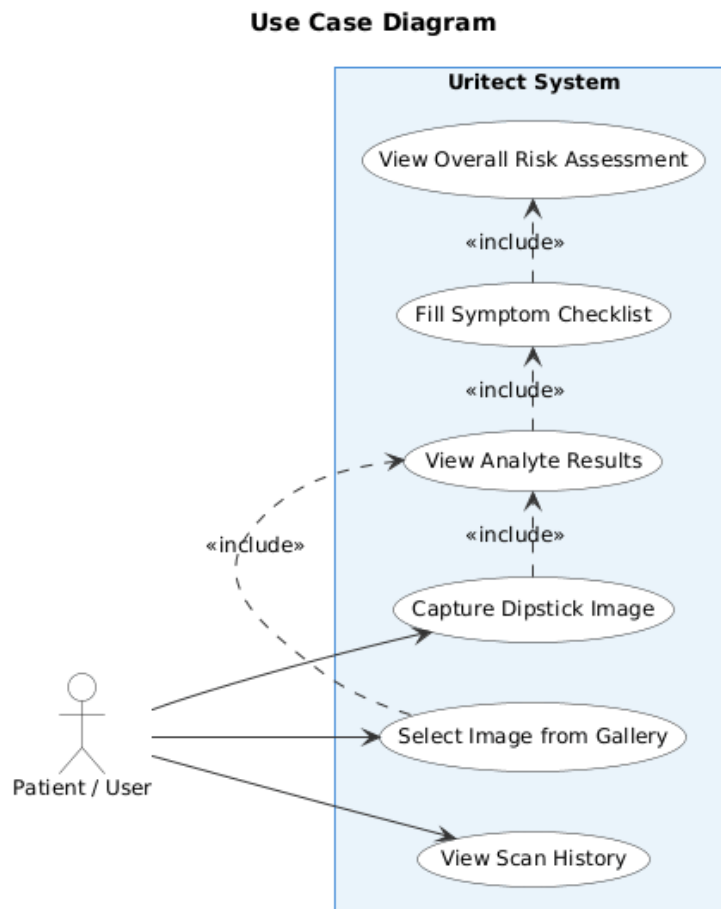


Fig. 8. Use Case Diagram

Figure 8 presents the use case diagram for Uritect with a single actor, Patient/User, reflecting the system's intended deployment for self-directed use at Rural Health Units without a distinct healthcare professional role. The actor initiates a scan through either Capture Dipstick Image or Select Image from Gallery, both of which include View Analyte Results, which in turn includes Fill Symptom Checklist, which includes View Overall Risk Assessment, forming a mandatory sequential chain. View Scan History is a direct, independent use case that does not require a new scan, allowing users to review previously saved records from the home page.

Activity Diagram

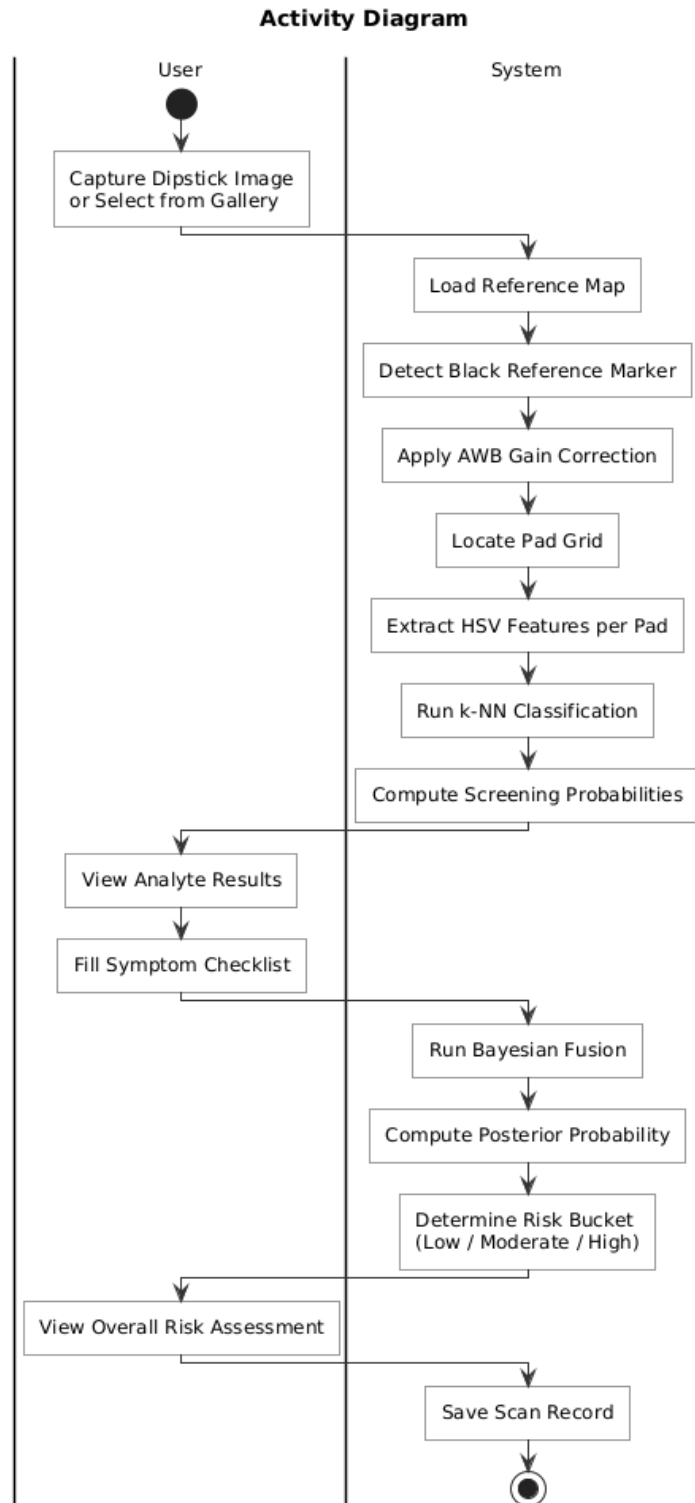


Fig. 9. Activity Diagram

Figure 9 presents the runtime activity of Uritect across two swimlanes: User and System. The user initiates the flow by capturing a dipstick image or selecting one from the gallery; from that point, the System lane executes all processing automatically – loading the reference map, detecting the black reference marker via flood-fill, applying AWB gain correction, locating the pad grid, extracting per-pad HSV features, running k-NN classification, and computing screening probabilities for the four sentinel analytes before returning control to the user to view analyte results. The user then fills the symptom checklist and submits it; the System lane runs Bayesian fusion, computes the posterior probability, determines the risk bucket, and saves the scan record as the final step before the user views the overall risk assessment.

Sequence Diagram

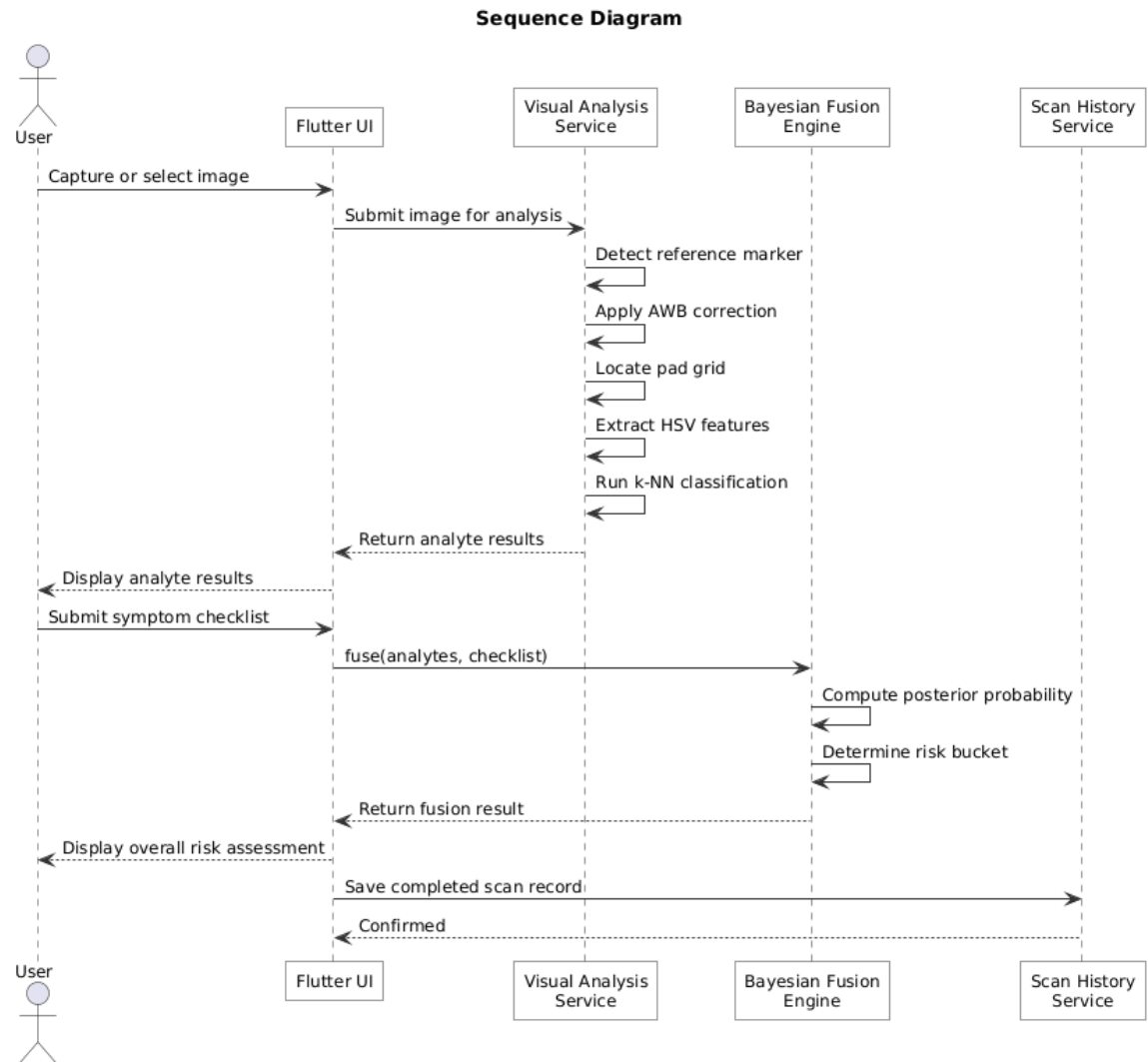


Fig. 10. Sequence Diagram

Figure 10 models the runtime message flow of Uritect across five participants: User, Flutter UI, Visual Analysis Service, Bayesian Fusion Engine, and Scan History Service. The user captures or selects an image; Flutter UI submits it to Visual Analysis Service, which executes the complete CV pipeline internally through self-calls for marker

detection, AWB correction, pad grid localization, HSV feature extraction, and k-NN classification, then returns the analyte results. Flutter UI displays these to the user, who then submits the symptom checklist; Flutter UI calls `fuse(analytes, checklist)` on the Bayesian Fusion Engine, which computes the posterior probability and determines the risk bucket before returning the fusion result. Flutter UI then dispatches a save to Scan History Service, which confirms completion; the overall risk assessment is shown to the user as the final step. There is no database participant because storage is a flat JSON file written via `path_provider`.

State Diagram

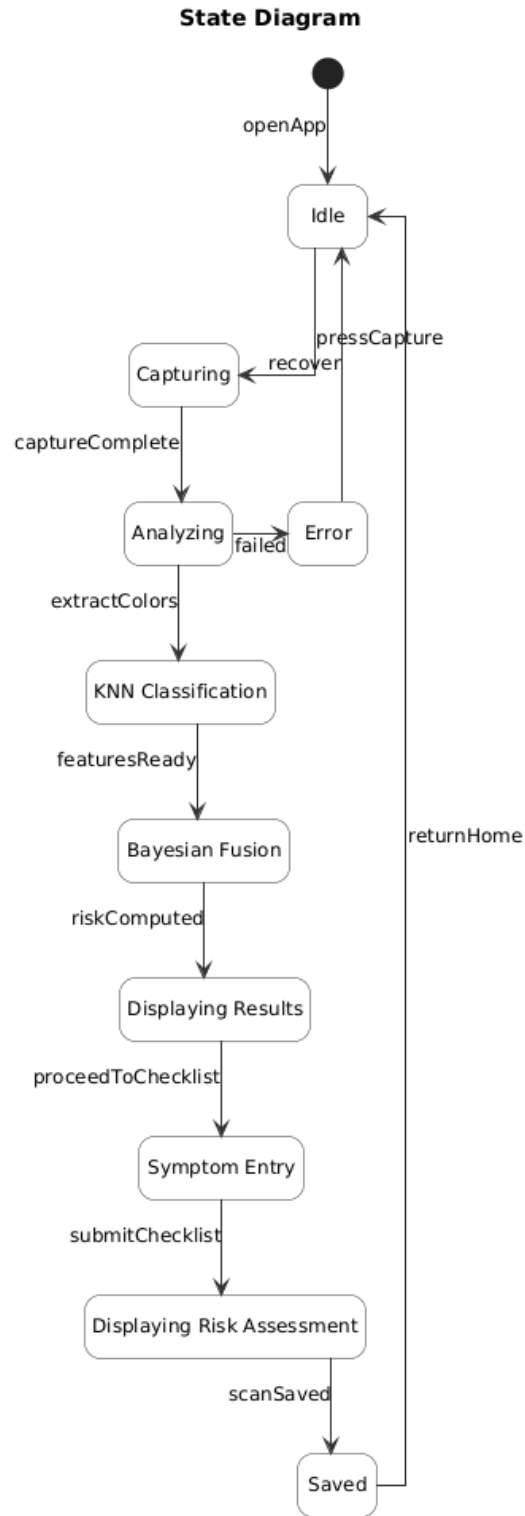


Fig. 11. State Diagram

Figure 11 models the application lifecycle of Uritect as a state machine. From Idle, the user presses capture to enter the Capturing state; completing the capture triggers the transition to Analyzing, from which a failure transitions to an Error state that recovers back to Idle. Successful analysis transitions through KNN Classification and into Bayesian Fusion, after which the application enters Displaying Results. The user proceeds to Symptom Entry and submits the checklist, transitioning to Displaying Risk Assessment; once the scan is saved the application enters Saved, from which returnHome transitions back to Idle for the next scan. The analysis begins automatically on image submission without any intermediate user confirmation, and the Error state handles only CV-level failures such as an undetectable marker.

Deployment Diagram

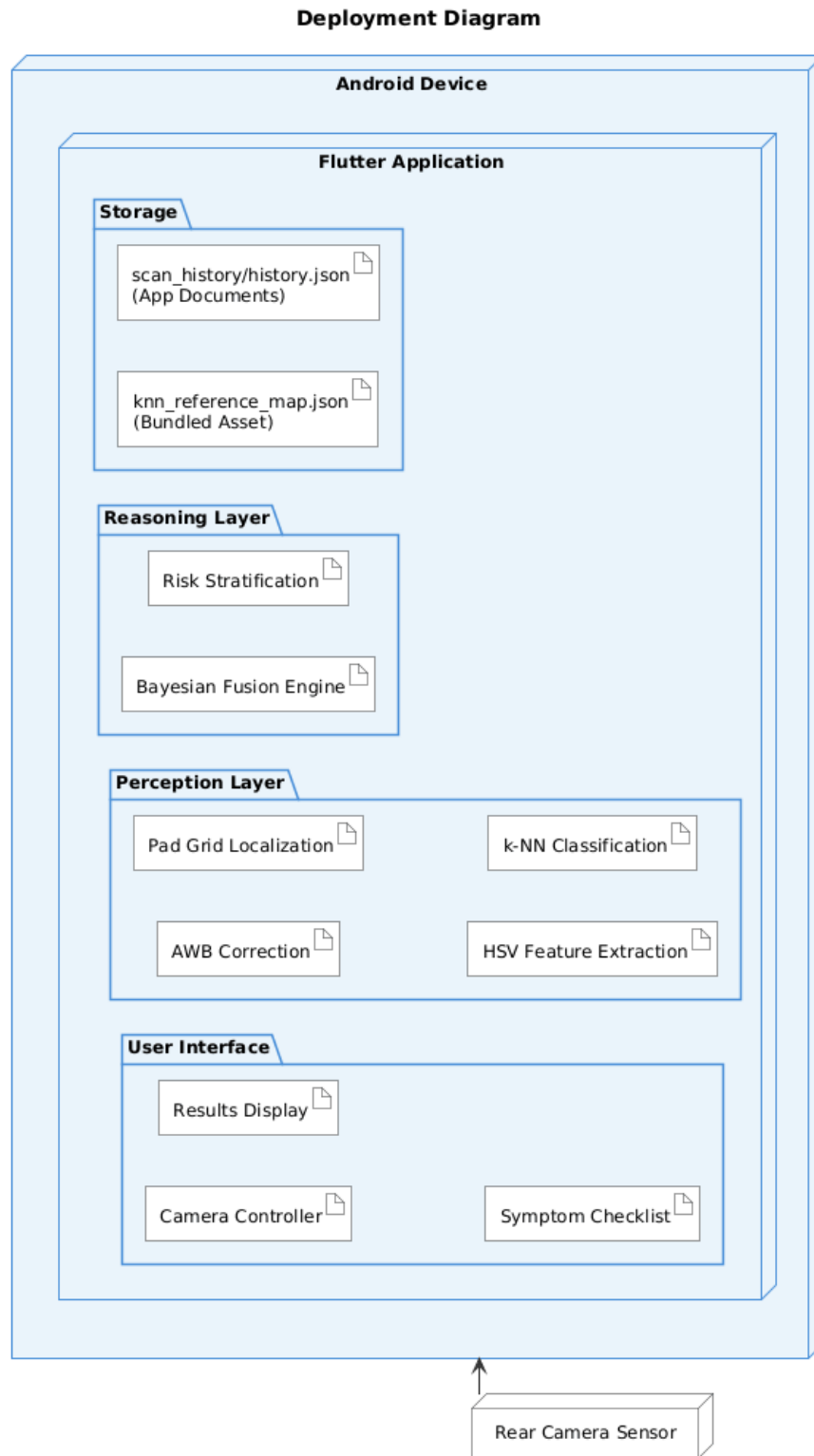


Fig. 12. Deployment Diagram

Figure 12 depicts the physical deployment of Uritect on a single Android device with no external server or internet dependency. The rear camera sensor feeds into the Android Device node, which hosts the Flutter Application partitioned into four packages: User Interface (Camera Controller, Results Display, Symptom Checklist), Perception Layer (AWB Correction, Pad Grid Localization, HSV Feature Extraction, k-NN Classification), Reasoning Layer (Bayesian Fusion Engine, Risk Stratification), and Storage (knn_reference_map.json as a bundled read-only asset and scan_history/history.json in the app documents directory). The camera sensor is the only external hardware dependency; all computation, all reference data, and all scan history reside within the device.

Process Design

LEVEL 0

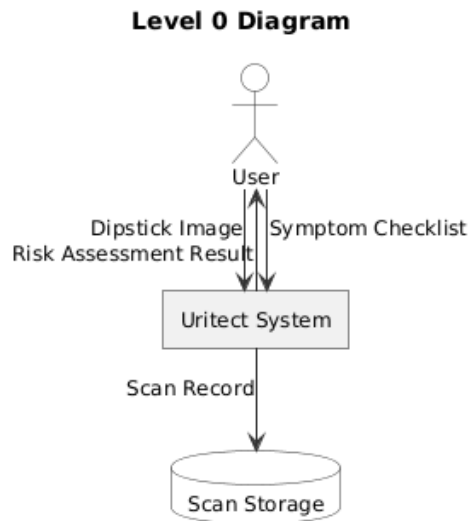


Fig. 13. Level 0 Diagram

Figure 13 presents the Level 0 data flow diagram, or context diagram, for Uritect. A single process bubble represents the entire system. The User external entity supplies two inputs: a Dipstick Image and a Symptom Checklist, and receives one output: the Risk Assessment Result. A second external entity, Scan Storage, receives the Scan Record produced after fusion completes. Image acquisition is internal to the system and is not represented as a separate external entity at this level of abstraction.

LEVEL 1

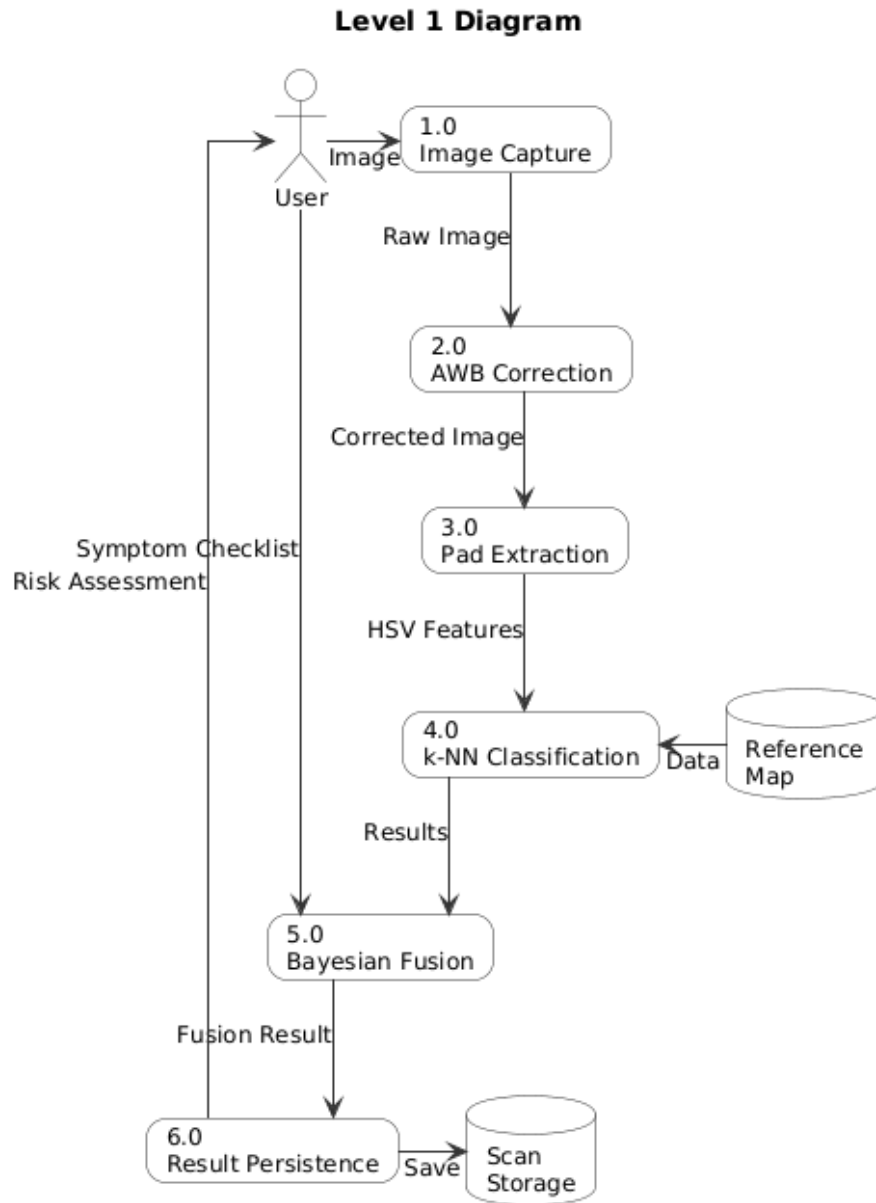


Fig. 14. Procedural Diagram

Figure 3.14 decomposes the Uritect system into six sequential processes. Process 1.0 (Image Capture) accepts the raw image from the User and passes it as a Raw Image to

2.0 (AWB Correction), which outputs a Corrected Image to 3.0 (Pad Extraction). Process 3.0 produces an HSV Features vector for each detected pad, feeding 4.0 (k-NN Classification), which reads reference data from the Reference Map data store – a read-only bundled JSON asset. Process 4.0 passes analyte results to 5.0 (Bayesian Fusion), which also receives the Symptom Checklist directly from the User; 5.0 produces a Fusion Result that 6.0 (Result Persistence) writes as a Scan Record to Scan Storage and returns the Risk Assessment to the User. The pipeline is strictly sequential with no parallel branches, and the Reference Map is a read-only store that is never written to on device.

System Development Life Cycle

To systematically develop, calibrate, and evaluate the offline Clinical Decision Support System (CDSS), this study adopts the Prototyping System Development Life Cycle (SDLC). This iterative methodology is specifically selected because the formulation of the system's reasoning layer (Bayesian Late Fusion) is strictly dependent on the empirical data and performance metrics derived during the calibration of the perception layer (k-NN visual

classification). Continuous refinement through successive prototyping ensures that algorithmic constraints—such as illumination robustness and edge-computing efficiency—are rigorously met before human sample validation.

The development process is structured into five iterative phases :

Phase 1: Requirements Gathering and Architectural Bounding

During this initial phase, the operational boundaries and technical constraints of the system are defined. This includes restricting the deployment architecture to an offline, Android-based edge-computing environment to ensure viability in geographically isolated Rural Health Units (RHUs). The illumination-adaptive parameters are bounded to a predefined color temperature range of 2700K (Warm White) to 5500K (Cool Daylight). Furthermore, the specific clinical targets for the Bayesian inference engine are restricted to Urinary Tract Infection (UTI) and predefined renal impairment indicators.

Phase 2: Initial Prototype Construction

The first working prototype focuses purely on the computer vision pipeline. The Android application is

programmed to capture dipstick images and execute the sequence of local image processing algorithms. This phase involves coding the Adaptive White-Balance (AWB) using the Reference-Based Gray World Assumption targeting the unreacted white plastic of the strip as the static reference followed by a marker-anchored brute-force grid search for pad localization. The prototype is engineered to successfully extract mean HSV (Hue, Saturation, Value) feature vectors from the 10 chemically treated reagent pads.

Phase 3: Iterative Calibration and Model Training

- This phase represents the core of the prototyping cycle, operating within a controlled laboratory environment. The k-NN classifier is trained on human urine specimens collected from healthy volunteers and processed by licensed medical technologists at the partner RHU, consistent with the data acquisition framework described.
- Images are continuously captured across the three predefined lighting conditions (2700K, 4000K, 5500K) for each volunteer specimen to generate a labeled colorimetric reference dataset.

- The k-Nearest Neighbors (k-NN) algorithm is trained on the extracted HSV vectors, with ground-truth labels assigned by the licensed medical technologist at the partner facility.
- If the classifier exhibits instability due to residual lighting noise, the prototype returns to Phase 2 for refinement of the AWB gain parameters and pad grid search parameters. This loop continues until colorimetric separability and visual probability outputs are stabilized.

Phase 4: Prototype Refinement and Bayesian Integration

Once the perception layer consistently outputs reliable soft visual probabilities, the prototype is expanded to include the Reasoning Layer. The clinical metadata interface is built, allowing the input of binary patient symptoms (e.g., dysuria, urgency, fever). The Bayesian Late Fusion engine is integrated directly onto the device, parameterized with fixed disease prevalence priors and established clinical likelihoods (e.g., the high specificity of the Nitrite parameter for UTI). The continuous posterior probabilities generated by the Bayes formulation are then mapped to the discrete risk stratification thresholds (Low, Moderate, High Risk).

Phase 5: System Deployment and Clinical Validation

The finalized, fully functional offline prototype is deployed for real-world validation in coordination with a partner healthcare facility. To establish a robust clinical ground truth in the absence of automated laboratory analyzers, the manual interpretation of the human urine samples during Phase 2 will not rely on a single reader. Instead, the study utilizes an Expert Consensus Panel. The ground truth for each real-world urine sample will be determined by the majority consensus of at least two independent, licensed Medical Technologists evaluating the dipstick under standardized, controlled lighting. This consensus approach significantly reduces the subjective inter-reader variability inherent to visual inspection, providing a mathematically stronger baseline for evaluating the system's accuracy and agreement. In this phase, algorithmic calibration ceases, and the system is evaluated strictly as a static screening tool. A recognized limitation of standard urinalysis is the subjectivity and high inter-reader variability of manual dipstick interpretation. Therefore, comparing the system's output against a single medical technologist's visual reading introduces a circular validation fallacy.

To establish a rigorous clinical ground truth without access to automated laboratory analyzers (such as the CLINITEK Novus), this study utilizes a Dual Expert Consensus as a surrogate gold standard. During the Phase 2 clinical trials, the baseline classification for every human urine sample will be determined by the blinded, independent evaluation of two licensed Medical Technologists. To establish the definitive ground truth, a 100% agreement must be reached between both professionals. In instances of discordant readings between the two technologists, a joint re-evaluation of the dipstick will be conducted to achieve a final consensus; if consensus cannot be reached, the sample will be excluded from the final validation dataset. The system's computer vision output will then be statistically compared against this strict, dual-verified ground truth using Cohen's Kappa coefficient to measure inter-rater reliability, ensuring a mathematically robust evaluation of the software's diagnostic accuracy.

Evaluation metrics, including Cohen's Kappa coefficient, Sensitivity, Specificity, and device computational efficiency (in milliseconds), are calculated to determine the success of the prototype.

3.2 Validation Protocol with Human Urine Samples

Following completion of algorithm development and internal performance evaluation, the system undergoes a structured validation phase at a licensed partner healthcare facility.

During this phase, routine dipstick urinalysis is performed on human urine specimens by licensed medical technologists as part of the facility's standard clinical workflow. The role of the researchers is limited to:

- Capturing standardized smartphone images of completed dipstick readings under ambient facility lighting
- Recording the corresponding manual interpretations independently provided by the medical technologist

The researchers do not participate in specimen collection, chemical processing, or clinical interpretation.

Human urine samples used in the validation phase are collected, handled, processed, and disposed of exclusively by licensed healthcare personnel at the partner facility in accordance with established laboratory protocols. Volunteer urine specimens used during the training phase described

were similarly forwarded to the partner RHU for processing under standard laboratory protocols. The researchers did not directly collect, store, transport, or manipulate biological specimens at any stage.

This validation phase does not constitute a clinical diagnostic trial. It does not seek to establish definitive population-level diagnostic accuracy, predictive values, or outcome-based effectiveness. Its methodological purpose is to evaluate agreement between:

- System-generated pad classifications
- System-generated posterior risk stratifications
- Manual visual interpretations by trained medical technologist under real-world ambient lighting conditions.

The validation is therefore framed as an inter-method concordance assessment rather than a clinical replacement evaluation.

Agreement Evaluation Metrics

System outputs are compared against medical technologist interpretations using the following metrics:

Sensitivity and Specificity

Sensitivity and specificity are computed per dipstick parameter and per disease risk category.

- Sensitivity – proportion of manually identified positive cases correctly identified by the system
 - Specificity – proportion of manually identified negative cases correctly identified by the system

These metrics are interpreted strictly as measures of directional agreement with manual reading, not as laboratory-standard diagnostic validation.

Overall Accuracy

Overall accuracy is reported as the proportion of instances in which the system classification agrees with the manual interpretation across all evaluated parameters.

Because accuracy does not account for chance agreement, it is interpreted as a supplementary descriptive metric rather than a primary agreement measure.

Diagnostic Performance Metrics

Due to the critical nature of clinical decision support, overall system accuracy is an insufficient metric. The system's diagnostic validity will be evaluated using a

standard clinical confusion matrix. The primary metrics extracted will include Sensitivity (the true positive rate, measuring the system's ability to correctly identify pathological analyte presence) and Specificity (the true negative rate, measuring the ability to correctly identify healthy samples). Additionally, the Positive Predictive Value (PPV) and Negative Predictive Value (NPV) will be calculated to assess the post-test probability of the system's outputs, ensuring the k-NN classifier provides statistically safe triage recommendations.

Cohen's Kappa Coefficient (κ)

Cohen's Kappa is used as the primary concordance metric, as it adjusts for agreement occurring by chance and is more appropriate than raw accuracy for categorical and ordinal classification comparison.

Interpretive thresholds follow commonly cited frameworks:

- 0.61 - 0.80 - Substantial agreement
- > 0.80 - Near-perfect agreement

Kappa is emphasized because the objective of validation is inter-method agreement rather than predictive modeling performance.

Cross-Device Generalization Protocol

To substantiate the claim that the system functions as a generalized 'Smartphone-Based' application, the validation protocol incorporates a Cross-Device Generalization Phase. Because different smartphone camera sensors and Image Signal Processors (ISPs) exhibit varying colorimetric responses, validating on a single device is insufficient. During Phase 2, the application's Adaptive White Balance (AWB) and k-NN classifier will be evaluated concurrently across three distinct mid-range Android devices representing different hardware manufacturers (e.g., Samsung, Xiaomi, and Oppo). System accuracy and precision will be measured on each device using the identical urine sample under the same lighting conditions. This multi-device evaluation will empirically demonstrate the AWB algorithm's capacity to standardize color variations across different camera hardware, proving the system's viability for widespread deployment in Resource-Constrained Health Units.

Software Quality Evaluation (ISO/IEC 25010)

Functional Suitability and Statistical Rigor

To ensure deployment suitability for Rural Health Units (RHUs), the system is evaluated against the ISO/IEC 25010 standard. The minimum sample size is formally calculated using the Hajian-Tilaki method for diagnostic studies to achieve a ± 0.07 half-width for the 95% confidence interval of analyte sensitivity. Functional suitability is established by measuring functional correctness, defined as the overall classification accuracy across all ten urinalysis parameters compared to a laboratory reference. Risk stratification evaluates the correctness of the Bayesian model in assigning low, moderate, and high risk levels; this is assessed using the Brier score for probability calibration and quadratically weighted Cohen's Kappa for ordinal agreement, requiring a minimum $\kappa \geq 0.60$. Statistical uncertainty for concordance metrics and the Brier score is estimated using bootstrap resampling with 1,000 iterations. McNemar's test is utilized for paired nominal data comparisons between the automated system and manual visual interpretations.

Performance Efficiency and System Provenance

Performance efficiency is quantified through latency and computational resource constraints necessary for offline edge-device operation. Time behavior is measured by

tracking the median and 95th-percentile processing latency from image capture to final risk output display, mandating an acceptance threshold of ≤ 5000 milliseconds for the 95th percentile. Resource utilization is monitored by recording peak RAM consumption and maximum CPU utilization during inference. The system must operate within a strict computational resource budget to guarantee application stability on entry-level Android hardware; monitoring ensures the application avoids memory overflow and thermal throttling on devices meeting the 4 GB RAM minimum baseline. To enable post-deployment auditing and reproducibility, all inference sessions are recorded in a local JSON file (scan_history/history.json) written to the application documents directory. Each saved scan record captures the full ScanResult including all ten analyte classifications, the four sentinel analyte screening probabilities, the symptom checklist selections, the clinical log-odds contribution, and the posterior probability and risk bucket, ensuring diagnostic outputs can be reviewed against the complete inference state that produced them..

Usability, Portability, and Generalization

Usability and portability are examined to validate environmental and hardware robustness. Illumination stability is quantified by computing the mean ΔE color difference and HSV variance of classification outputs across the 2700 K to 5500 K lighting range. A successful AWB implementation requires an empirically chosen target of mean $\Delta E < 5.0$. Cross-device generalization is tested by sampling images across three distinct smartphone models representing entry-level, mid-range, and flagship tiers; hardware-induced classification variance is statistically analyzed to isolate device-specific performance drops. User acceptance is measured via the Technology Acceptance Model (TAM) using standardized surveys to quantify perceived usefulness and ease of use among target RHU healthcare workers.

Ethical and Data Handling Considerations

This study adheres to the ethical principles governing research involving human biological specimens and healthcare data. Human urine samples used in the validation phase are collected, handled, and interpreted exclusively by licensed healthcare personnel at the partner facility as

part of established routine clinical procedures. The researchers do not have direct access to patient identifiers, medical records, or biological specimens at any point.

System-generated outputs from the validation phase are recorded in anonymized or de-identified form, with no linkage to individual patient identities. Image data captured during validation is used solely for classification agreement analysis and is not retained beyond the purposes of this study. The study does not modify clinical decision-making for any patient, as the system operates in a parallel observation capacity during the validation phase without influencing the medical technologist's independent manual interpretation.

The system's outputs are clearly framed within the application interface as preliminary risk stratification only. Users are explicitly directed to seek confirmation from a licensed physician or healthcare facility regardless of the system's risk classification output, consistent with the CDSS scope defined in Chapter 1.

CHAPTER 4 RESULTS AND DISCUSSION

Implementation

The Uritect system was implemented as an Android-first mobile application utilizing Flutter/Dart-based image interpretation workflows supported by Python-based preprocessing experimentation and evaluation tools. The current implementation focuses on semiquantitative screening support through a guided workflow combining smartphone-based image acquisition, image preprocessing, analyte segmentation, HSV-based feature extraction, semiquantitative analyte interpretation, and structured result presentation within a unified mobile application environment.

The implemented workflow begins with dipstick image capture using a smartphone camera and proceeds through multiple image-processing stages, including orientation handling, marker-based localization, image normalization, reagent-pad segmentation, HSV-based feature extraction, and k-Nearest Neighbors (k-NN)-based analyte matching. The extracted analyte-level information is then presented through a structured clinical-style interface intended to

support preliminary urinalysis interpretation and non-diagnostic interpretation decision support.

The implemented system processes all ten reagent-strip analytes during interpretation, namely Leukocytes/Leukocyte Esterase, Nitrite, Urobilinogen, Protein, pH, Blood, Specific Gravity, Ketone, Bilirubin, and Glucose. In addition, the application integrates symptom-assisted Bayesian fusion logic to support risk-oriented screening interpretation for clinically prioritized analytes associated with urinary tract infection and renal-risk screening.

The application workflow follows the sequence:

Home Page → Capture Dipstick Page → Analyzing Page → Results Page → Symptom Checklist Page → Overall Results Page

The workflow was designed to improve interpretation organization, support smartphone-assisted urinalysis screening, and provide a simplified mobile interaction process within a lightweight offline-capable environment. The implemented workflow additionally supports local scan history storage and structured result presentation to improve usability during repeated screening sessions.

Development Tools

The development of the Uritect system required the integration of multiple development environments, frameworks, libraries, and supporting tools for mobile application implementation, image-processing experimentation, and deployment testing. The selected technologies were chosen based on compatibility with Android development, suitability for computer vision preprocessing, offline execution capability, and ease of iterative experimentation during implementation.

TABLE V DEVELOPMENT TOOLS AND TECHNOLOGIES

Category	Tool/Specification	Purpose	Reason for Selection
Operating Environment	Windows OS	Main development environment	Compatible with Flutter, Android SDK, and Python-based workflows
Mobile Development Framework	Flutter	Mobile application development	Supports rapid UI development and Android-first deployment
Programming Language	Dart	Flutter application development	Native language support for Flutter
IDE / Editor	Visual Studio Code	Source code editing and debugging	Lightweight environment with Flutter and Python extensions

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Android Development Environment	Android Studio	Emulator testing and SDK integration	Official Android development platform
Image Processing Environment	Python 3	Pipeline implementation and experimentation	Suitable for scientific computing and preprocessing workflows
Computer Vision Library	OpenCV	Image preprocessing and segmentation	Optimized for computer vision operations
Numerical Computing Library	NumPy	Numerical computations and feature handling	Efficient matrix and array operations
Version Control	Git / GitHub	Version management and backup	Supports iterative development and revision tracking
UI/UX Prototyping	Figma	Interface prototyping and layout planning	Rapid interface visualization and workflow design
Design Support	Canva	Visual asset preparation	Used for interface assets and presentation materials

Flutter and Dart were utilized for Android application development due to their support for rapid interface prototyping and modular mobile application architecture. Visual Studio Code and Android Studio supported source-code editing, SDK integration, emulator testing, deployment, and

debugging workflows throughout implementation. Python 3 together with OpenCV and NumPy supported preprocessing experimentation, HSV feature extraction, and semiquantitative analyte evaluation during development and dataset analysis.

TABLE VI PYTHON LIBRARIES UTILIZED

Library	Version / Requirement Used	Purpose in Uritect
Pillow (PIL)	≥10.0.0	Used for image loading, EXIF orientation correction, and image decoding during preprocessing and dataset preparation.
pillow-heif	≥0.16.0	Enables loading and conversion of HEIC image formats during dataset preparation.
NumPy	≥1.24.0	Performs numerical array processing for HSV feature extraction, analyte handling, and interpretation-related computations.
pandas	≥2.0.0	Supports dataset organization and structured tabular data handling during experimentation and preprocessing.
scikit-learn	≥1.3.0	Implements k-Nearest Neighbors (k-NN) classification, train-test splitting, confusion matrix generation, and evaluation metrics including accuracy, precision, recall, F1-score, and Cohen's kappa.

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imbalanced-learn	≥0.11.0	Provides SMOTE-based oversampling support for handling class imbalance during model experimentation.
OpenCV (cv2)	≥4.10.0	Performs computer vision operations including image conversion, contour detection, thresholding, perspective transformation, and analyte localization during preprocessing experimentation.
FastAPI	≥0.100.0	Provides the local scan-server API used during development and testing workflows.
uvicorn[standard]	≥0.22.0	Runs the FastAPI local development server during preprocessing experimentation.
python-multipart	≥0.0.6	Handles multipart image upload processing within the FastAPI scan-server workflow.
SciPy		Used for logistic and Bayesian fusion computations through <code>scipy.special.expit</code> .

Python standard libraries such as `csv`, `json`, `math`, `pathlib`, `statistics`, `datetime`, `asyncio`, `pickle`, and `uuid` were additionally utilized throughout preprocessing, local storage handling, dataset management, and interpretation workflows. Since these libraries are included within the Python standard library, separate dependency versions were not required.

TABLE VII DART/FLUTTER LIBRARIES UTILIZED

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Library / Package	Pubspec Constraint	Resolved Version	Purpose in Uritect
Dart SDK	^3.9.2	≥3.9.2 <4.0.0	Core programming language and runtime environment used throughout the application.
Flutter SDK	SDK dependency	≥3.35.0	Main framework used for Android mobile application development and interface rendering.
flutter/material.dart	Flutter SDK	SDK	Provides Material Design widgets, layouts, navigation, and theming components.
flutter/services.dart	Flutter SDK	SDK	Loads bundled application assets such as the k-NN reference map JSON file.
flutter_test	Flutter SDK	SDK	Supports widget and interface testing during development.
cupertino_icons	^1.0.8	1.0.8	Provides additional icon resources used within the interface.
google_fonts	^7.0.0	7.1.0	Implements the Poppins and Inter typography used throughout the application interface.
camera	^0.10.5+5	0.10.6	Handles camera initialization, preview rendering, flash control, and dipstick image capture.

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image_picker	^1.0.7	1.2.2	Allows image selection from the smartphone gallery during testing and acquisition workflows.
http	^1.1.0	1.6.0	Supports communication with the optional local Python scan server during development experimentation.
image	^4.8.0	4.8.0	Performs on-device image decoding, resizing, HSV computation, and analyte feature extraction within the Flutter application.
path_provider	^2.1.5	2.1.5	Retrieves local application storage paths for scan history and persisted scan records.
flutter_lints	^5.0.0	5.0.0	Provides static analysis and linting support during development.

Dart built-in libraries including `dart:convert`, `dart:io`, `dart:async`, `dart:isolate`, and `dart:math` were additionally utilized throughout the implementation of local storage handling, asynchronous processing, JSON serialization, isolate-based processing, and numerical operations within the mobile application workflow.

Hardware Requirements

The implementation, testing, and deployment of the Uritect system required hardware capable of supporting mobile application execution, image-processing operations, and smartphone-based image acquisition. Since the proposed study combines a Flutter-based Android application with a Python/OpenCV image-processing pipeline, the hardware requirements were categorized into development and testing specifications and deployment-oriented mobile device requirements.

The selected hardware configurations were intended to support stable image acquisition, preprocessing operations, and semiquantitative interpretation during implementation and testing.

Hardware Used for Development and Testing

The development and testing phase required hardware capable of supporting Flutter application development, Android emulator execution, and image-processing experimentation.

TABLE VIII HARDWARE SPECIFICATIONS FOR THE MOBILE
APPLICATION DEVELOPMENT

Component	Specification	Purpose
Device Name	SKP-00	Main development environment
Processor	AMD Ryzen 5 5500U with Radeon Graphics (2.10 GHz)	Mobile application development and image-processing operations
RAM	8 GB RAM (7.35 GB usable)	Multitasking, emulator execution
Graphics	AMD Radeon Graphics	UI rendering and visualization
Storage	477 GB SSD Storage	Dataset, project files, and software installation
System Type	64-bit operating system, x64-based processor	Compatibility with Android and Python environments

The AMD Ryzen 5 5500U processor and 8 GB RAM configuration provided sufficient capability for Flutter development, OpenCV preprocessing, and emulator execution. SSD-based storage additionally improved application responsiveness and dataset access during implementation.

During experimentation, repeated preprocessing operations occasionally increased memory utilization, particularly during multi-frame image-processing workflows.

Mobile Device Testing Specifications

TABLE IX MOBILE DEVICE TESTING SPECIFICATIONS

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Component	Specification	Purpose
Device Name	OPPO Reno8 Z 5G	Mobile application testing
Model	CPH2457	Device identification
Processor	Qualcomm Snapdragon 695 5G Octa-core	Real-device application execution
RAM	8 GB RAM	Mobile multitasking and processing stability
Storage	128 GB Storage	Application and image storage
Battery Capacity	4500 mAh	Extended testing sessions
Screen Size	6.43-inch Display	User interface and workflow evaluation
Android Version	Android 14	Compatibility testing
Operating Environment	ColorOS 14	Runtime environment validation

The OPPO Reno8 Z 5G device was utilized to evaluate the implemented application under real-device conditions. Testing focused on navigation consistency, image acquisition behavior, preprocessing stability, and deployment-oriented operation.

The Android 14 environment supported runtime compatibility testing for camera integration and image acquisition workflows. Minor inconsistencies in automatic

exposure and image enhancement behavior were observed during testing.

Image Acquisition Device

TABLE X IMAGE ACQUISITION DEVICE SPECIFICATIONS

Component	Specification	Purpose
Device Name	Xiaomi 11T	Dipstick image acquisition
Rear Camera System	108MP Main: \ (1/1.52" \) sensor, 0.7µm pixels (12MP output with 9-in-1 Super Pixel), f/1.75 aperture, PDAF (no OIS)	Urinalysis dipstick capture
Operating Environment	Android-based smartphone	Real-world imaging evaluation

The Xiaomi 11T smartphone was primarily used for dipstick image acquisition during implementation and testing. The device's rear camera system supported improved analyte visibility, and clearer image detail retention.

Mobile Device Hardware Requirements

The Uritect system was designed to operate on Android smartphones capable of supporting Flutter-based application execution, smartphone imaging, and lightweight image-processing operations.

TABLE XI MINIMUM MOBILE DEVICE REQUIREMENTS

Component	Minimum Requirement
Processor	Quad-core mobile processor
RAM	4 GB
Storage	4 GB available storage
Display	720p display
Rear Camera	8 MP smartphone camera
Android Version	Android 10 or later

TABLE XII RECOMMENDED MOBILE DEVICE REQUIREMENTS

Component	Recommended Requirement
Processor	Octa-core mobile processor
RAM	6 GB or higher
Storage	8 GB available storage
Display	1080p display
Rear Camera	12 MP or higher smartphone camera
Stable Stand/Tripod	Recommended for image stability

The proposed hardware requirements were selected to support stable image acquisition, preprocessing operations, and semiquantitative interpretation within Android-based environments. Devices with higher RAM capacity and improved

camera systems demonstrated more stable preprocessing performance and analyte visibility during testing.

Although the implemented preprocessing workflow reduced several image acquisition inconsistencies, uncontrolled environmental conditions occasionally affected analyte representation during operation.

Software Requirements

The implementation of the Uritect system required multiple software environments, frameworks, libraries, and runtime components to support mobile application development, preprocessing operations, experimentation, and deployment.

TABLE XIII SOFTWARE REQUIREMENTS

Software	Purpose	Requirement
Windows OS	Main development environment	Windows 10 or later
Flutter SDK	Mobile application development	Latest stable version
Dart SDK	Flutter programming language support	Compatible with Flutter SDK
Android Studio	Android SDK integration and emulator testing	Latest stable version
Android SDK	Android application build and deployment support	Android 10 or later

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Visual Studio Code	Source code editing and debugging	Latest stable version
Python 3.x	Image-processing pipeline implementation	Python 3.10 or later
OpenCV	Computer vision and preprocessing operations	OpenCV-Python package
NumPy	Numerical processing and feature handling	Latest compatible version
Git / GitHub	Version control and collaboration	Optional but recommended
pip / Virtual Environment	Python package management	Required
Android Runtime Permissions	Camera and storage access	Required during deployment

The software stack enabled integration between the Android application layer and the image-processing environment utilized during experimentation and evaluation. Flutter and Dart supported implementation of the Android-first mobile application, while Python 3, OpenCV, and NumPy supported preprocessing, segmentation, and feature extraction within the computer vision pipeline.

Android Studio and the Android SDK were utilized for emulator testing, deployment validation, and runtime compatibility evaluation. Additional tools such as pip, virtual environments, and Git/GitHub supported dependency management and version tracking throughout implementation.

The overall software architecture was intentionally designed to support lightweight local execution without dependence on cloud-based processing services.

Input and Outputs of the Study

This section presents the implemented interfaces of the Uritect mobile application and describes the corresponding inputs, processing stages, and generated outputs within the current prototype workflow. The application was designed to guide users through dipstick image acquisition, preprocessing, semiquantitative interpretation, and preliminary interpretation result presentation using a structured Android-based interface.

The implemented workflow currently consists of the Welcome Screen, Home Page, Capture Dipstick Page, Analyzing Page, Dipstick Results Page, and Overall Results Page. Each interface contributes to the overall smartphone-assisted urinalysis screening workflow of the application.



Fig. 15. Welcome Screen

The Welcome Screen serves as the initial entry interface of the Uritect application. The screen displays the system logo, branding elements, and an “Enter” button used to proceed to the main application workflow.

The primary input in this interface is the user interaction with the navigation button. The screen functions as an introductory interface intended to

establish application identity and provide an organized transition into the main workflow.

The output of this stage is the initialization of the application session and redirection to the succeeding application pages. The simplified layout and minimal interface elements were intentionally designed to reduce user confusion during application startup.

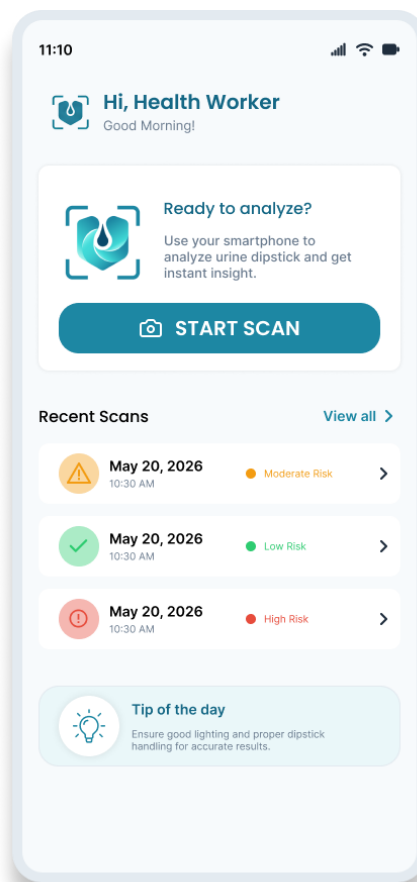


Fig. 16. Home Page

The Home Page functions as the primary navigation interface of the current application workflow. This page

serves as the entry point for initiating dipstick analysis and accessing implemented application features.

The interface currently displays the application branding and navigation controls that allow the user to proceed to the dipstick capture workflow. User interaction with the available controls serves as the primary input at this stage.

No preprocessing or analyte interpretation operations occur within this interface. Instead, the screen organizes the transition between the introductory workflow and the dipstick acquisition stage of the application.

The output of this stage is the initiation of the image acquisition workflow through user-directed navigation within the application environment.



Fig. 17. Capture Dipstick Page

The Capture Dipstick Page is responsible for dipstick image acquisition using the smartphone camera during the screening workflow. The interface allows users to capture reagent-strip images using the device camera prior to image interpretation and analyte processing.

The primary input at this stage is the captured dipstick image acquired using the smartphone's rear camera. The interface includes navigation controls and image-capture functions to support smartphone-based acquisition during

screening. Proper image capture at this stage is important because unstable positioning, glare reflections, shadows, and inconsistent lighting conditions may affect analyte visibility and HSV feature consistency during interpretation.

Once the image is captured, the system prepares the acquired image for image-processing operations, including orientation handling, marker-based localization, image normalization, reagent-pad segmentation, and HSV-based feature extraction.

The output of this stage is the captured dipstick image that serves as the primary input to the analyte interpretation workflow implemented within the mobile application.

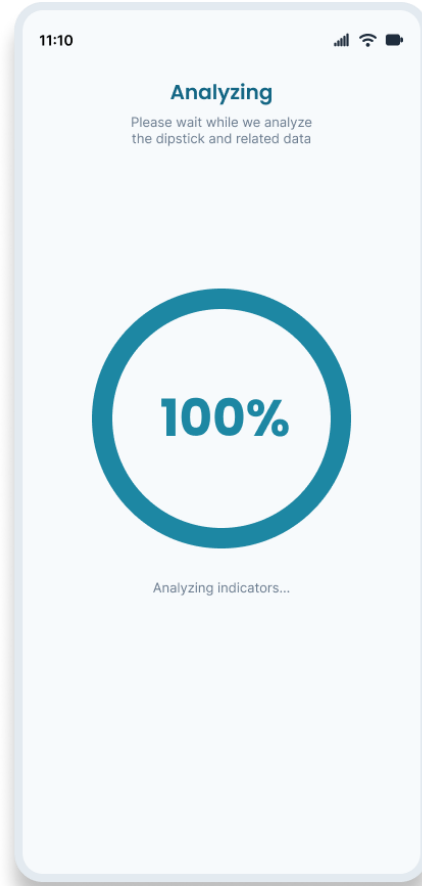


Fig. 18. Analyzing Page

The Analyzing Page presents the processing and interpretation stage of the application workflow. The interface displays a progress-oriented loading screen indicating that dipstick analysis and analyte interpretation operations are currently being performed.

The primary input at this stage is the captured dipstick image obtained from the image acquisition workflow. During this stage, the system performs image-processing and

analyte interpretation operations, including orientation handling, marker-based localization, reagent-pad segmentation, HSV-based feature extraction, and k-Nearest Neighbors (k-NN)-based analyte matching.

The displayed progress indicator functions as visual feedback to inform users that image interpretation and screening operations are ongoing. This interface was implemented to reduce uncertainty during processing and improve workflow continuity during interpretation execution.

The output of this stage is the completion of analyte-level interpretation and preparation of semiquantitative screening results for presentation within the succeeding workflow screens.



Fig. 19. Dipstick Results Page

The Results Page presents the semiquantitative analyte-level outputs generated by the implemented HSV-based interpretation workflow. The interface displays the detected analytes together with corresponding interpretation results and reference ranges for initial assessment presentation.

The primary input to this stage is the processed analyte-level feature information generated from the segmented

reagent pads. Following HSV-based feature extraction and k-NN analyte matching, the system maps the extracted features into semiquantitative analyte interpretations for presentation within the results interface.

The displayed output includes the following analytes:

- Leukocytes / Leukocyte Esterase
- Nitrite
- Urobilinogen
- Protein
- pH
- Blood
- Specific Gravity
- Ketones
- Bilirubin
- Glucose

Each analyte entry contains the detected interpretation result together with the corresponding reference range for interpretation support comparison. The interface additionally includes navigation controls for initiating symptom-assisted screening workflows and performing new dipstick scans.

The structured tabular layout was implemented to improve readability, analyte organization, and interpretation clarity during smartphone-assisted urinalysis screening.

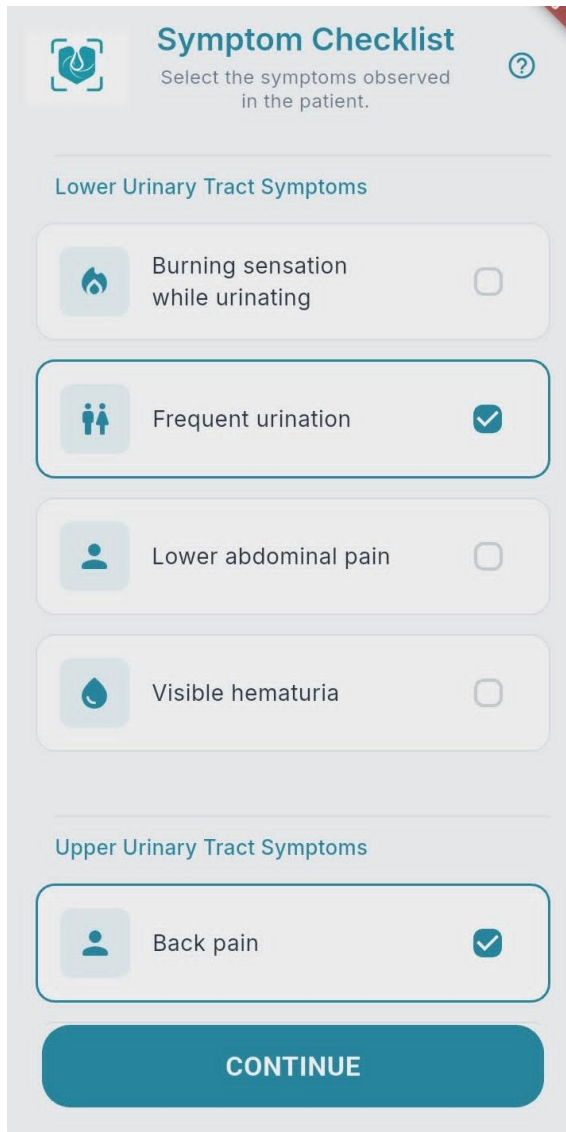
The image shows a mobile application interface for a 'Symptom Checklist'. At the top, there is a header with a heart icon inside a square, the title 'Symptom Checklist', and a subtitle 'Select the symptoms observed in the patient.' with a question mark icon. Below the header, the form is organized into two sections: 'Lower Urinary Tract Symptoms' and 'Upper Urinary Tract Symptoms'. Each section contains a list of symptoms with corresponding icons and checkboxes. In the 'Lower Urinary Tract Symptoms' section, 'Burning sensation while urinating' has an unchecked checkbox, 'Frequent urination' has a checked checkbox, 'Lower abdominal pain' has an unchecked checkbox, and 'Visible hematuria' has an unchecked checkbox. In the 'Upper Urinary Tract Symptoms' section, 'Back pain' has a checked checkbox. At the bottom of the form is a large blue button labeled 'CONTINUE'.

Fig. 20. Symptom Checklist Page

The Symptom Checklist Page allows users to input clinically relevant symptoms associated with urinary tract infection and renal-risk screening within the implemented

mobile workflow. The interface organizes symptoms into categories such as lower urinary tract symptoms and upper urinary tract symptoms to improve readability and simplify user interaction during screening.

The primary input at this stage consists of user-selected symptom information through interactive checklist controls. The implemented interface allows users to indicate symptoms including burning sensation while urinating, frequent urination, lower abdominal pain, visible hematuria, back pain, and other related urinary manifestations. Selected symptoms are visually highlighted within the interface to improve interaction clarity and screening continuity.

The collected symptom information is integrated with analyte-level interpretation outputs through the implemented Bayesian fusion workflow to support risk-oriented screening interpretation. The symptom-assisted workflow was designed to provide additional contextual support during preliminary urinalysis interpretation, particularly for analytes associated with urinary tract infection and renal-related screening conditions.

The generated output from this stage contributes to the computation of overall screening-oriented risk levels

presented within the Overall Results workflow. The implemented symptom-assisted reasoning module supports contextual interpretation while maintaining a simplified and structured mobile interaction process appropriate for implementation-stage screening support.

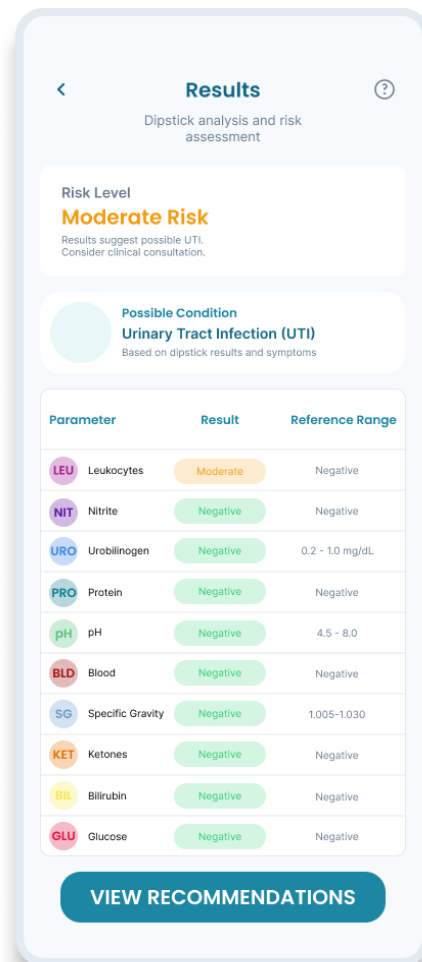


Fig. 21. Overall Results Page

The Overall Results Page presents the consolidated risk-based interpretation generated by the implemented mobile workflow. This page combines analyte-level findings,

symptom-assisted screening outputs, and risk-oriented interpretation presentation within a single clinical-style interface.

The primary input to this stage consists of analyte-level interpretation data generated from the HSV-based interpretation workflow, together with user-selected symptom information collected from the Symptom Checklist Page. The interface summarizes the interpreted dipstick findings and presents generalized risk-based interpretation outputs based on the currently implemented Bayesian fusion workflow.

The displayed output includes:

- detected risk level,
- possible condition indication,
- analyte-level interpretation summary,
- and navigation controls for viewing additional recommendations.

The interface integrates analyte-level interpretation findings with symptom-assisted Bayesian screening logic to generate preliminary risk-oriented outputs. The generated results are intended solely for assistive interpretation and should not be interpreted as definitive medical diagnoses.

The implemented workflow was designed to provide simplified mobile clinical-style interpretation support while maintaining lightweight offline-capable operation during smartphone-assisted urinalysis screening.

Results Interpretation and Analysis

This section presents the findings obtained during the implementation, testing, and evaluation of the current Uritect mobile application. The discussion focuses on the observed behavior of the implemented HSV-based image interpretation workflow, semiquantitative analyte matching, symptom-assisted screening logic, and practical findings identified throughout testing and experimentation.

The presented findings reflect the current implementation-stage evaluation results obtained during ongoing system refinement and dataset expansion. The implemented workflow is intended to support preliminary urinalysis screening and interpretation assistance within a mobile environment and is not intended to function as a standalone diagnostic system.

Image Interpretation and Feature Extraction Findings

The implemented Uritect workflow demonstrated the practical feasibility of performing smartphone-assisted semiquantitative urinalysis interpretation using a Flutter/Dart-based mobile application environment. The current implementation utilizes single-image capture, HSV-based feature extraction, k-Nearest Neighbors (k-NN)

analyte matching, and symptom-assisted Bayesian fusion to generate assessment support interpretation outputs.

During testing, image quality and environmental lighting conditions significantly affected analyte interpretation consistency. Variations in smartphone exposure behavior, mixed indoor lighting conditions, glare reflections, and inconsistent strip positioning occasionally altered analyte coloration and affected HSV feature consistency during interpretation.

The implemented HSV-based interpretation workflow demonstrated practical suitability for reagent-strip analysis because urine dipstick interpretation fundamentally relies on analyte color variation and gradual semiquantitative transitions between reaction levels. HSV representation was selected because it separates hue and saturation information from value or brightness information, making it more suitable than direct RGB values for colorimetric reagent-strip interpretation under variable illumination conditions.

The current implementation processes all ten reagent-strip analytes during interpretation, including Leukocytes/Leukocyte Esterase, Nitrite, Urobilinogen,

Protein, pH, Blood, Specific Gravity, Ketone, Bilirubin, and Glucose. However, the symptom-assisted screening and Bayesian fusion workflow prioritizes four clinically relevant analytes – Leukocytes/Leukocyte Esterase, Nitrite, Protein, and Glucose – because of their direct relevance to urinary tract infection and renal-risk screening objectives established in the study.

The implemented workflow additionally integrates symptom-assisted screening logic within the mobile urinalysis environment. Bayesian fusion combines semiquantitative analyte interpretation with user-provided symptom information to support risk-oriented screening outputs within the Overall Results workflow.

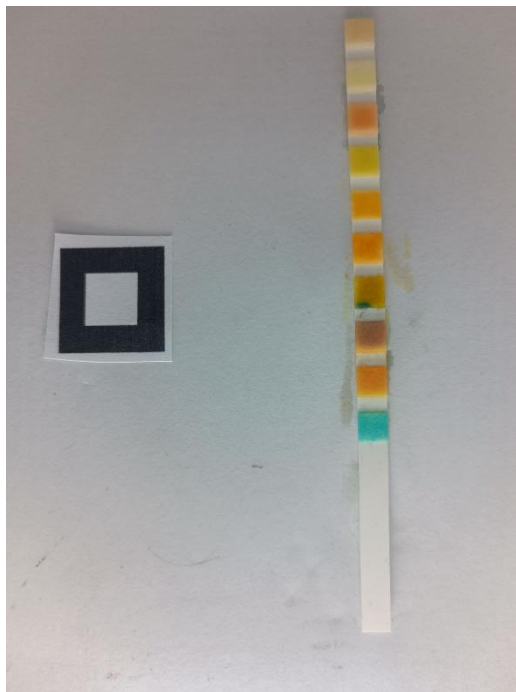


Fig. 22. Example dipstick image captured using the implemented smartphone-based acquisition workflow

The captured image demonstrates the single-image acquisition workflow currently implemented within the mobile application. Variations in illumination, glare reflections, and smartphone exposure behavior during acquisition occasionally affected analyte visibility and HSV feature consistency during interpretation.

TABLE XIV EXAMPLE HSV FEATURE REPRESENTATION EXTRACTED BY THE URITECT PIPELINE

Analyte	Mean H	Mean S	Mean V
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West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
Ia Paz, Iloilo City, Philippines

Leukocytes /Leukocyte Esterase	157.207199	0.032911	0.678400
Nitrite	59.148800	0.117749	0.657722
Urobilinogen	249.650406	0.030467	0.704949
Protein	238.369598	0.216166	0.982326
pH	272.798401	0.199652	0.654809
Blood	278.394409	0.291173	0.600202
Specific Gravity	239.975998	0.187283	0.525735
Ketone	248.902405	0.258458	0.873710
Bilirubin	248.726395	0.226202	0.878245
Glucose	88.258400	0.300427	0.649095

Values were extracted from the Uritet features dataset (features.csv) using the sample event:

BATCH_2_COOL_Abnormal_2026-03-19T21-03-34-843695

Hue values are represented in degrees, while saturation and value are normalized numerical color features used during k-NN reference-map comparison and semiquantitative analyte interpretation.

The extracted HSV feature values demonstrate how the implemented workflow converts reagent-pad coloration into analyte-level numerical representations for

semiquantitative interpretation. Variations in hue, saturation, and value measurements across analytes reflect the differing colorimetric behavior of each reagent pad during urine-strip analysis. These extracted features serve as the primary inputs to the k-NN-based analyte matching workflow implemented within the current Uritect system.

Performance Evaluation Results

The performance evaluation of the current Uritect implementation focused on semiquantitative analyte interpretation using HSV-based feature extraction, k-Nearest Neighbors (k-NN) classification, and preliminary symptom-assisted risk stratification within the implemented mobile interpretation workflow.

The implemented system processes all ten reagent-strip analytes during image interpretation and semiquantitative analysis. However, the symptom-assisted reasoning and risk stratification workflow currently prioritizes four clinically relevant analytes – Glucose, Protein, Leukocyte Esterase, and Nitrite – due to their direct association with urinary tract infection and renal-risk screening objectives established in the study.

Evaluation was conducted using overall accuracy, macro precision, macro recall, macro F1-score, weighted F1-score, and Cohen's kappa coefficient. These metrics were selected to evaluate both overall analyte interpretation consistency and multiclass semiquantitative performance across varying analyte categories and reaction levels.

TABLE XV PERFORMANCE EVALUATION METRICS

Metric	Value
Accuracy	52.47%
Macro Precision	19.74%
Macro Recall	22.27%
Macro F1-Score	19.20%
Weighted F1-Score	47.68%
Cohen's Kappa	50.14%

Performance Evaluation Metrics of the Uritect Interpretation Workflow

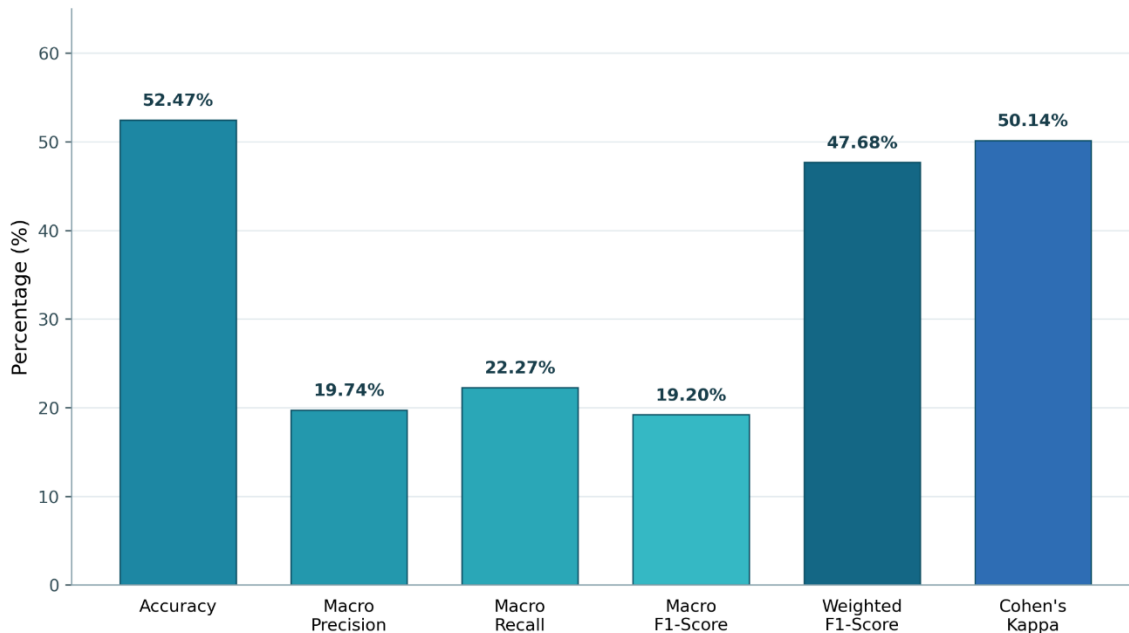


Fig. 23. Performance Comparison of the Current Semiquantitative Interpretation Workflow Across Multiple Evaluation Metrics

The current implementation achieved an overall accuracy of 52.47%, indicating moderate analyte-level interpretation performance within the evaluated dataset. The weighted F1-score of 47.68% similarly suggests moderate overall classification consistency when accounting for class distribution.

The obtained macro precision, macro recall, and macro F1-score values were comparatively lower than the weighted evaluation metrics. This behavior suggests that interpretation consistency varied across analyte categories

and semiquantitative levels. Since the dataset contains multiple analyte classes with potentially uneven distributions, macro-averaged metrics become particularly important for evaluating performance across all analyte categories rather than emphasizing dominant classes alone.

The obtained Cohen's kappa value of 50.14% indicates moderate agreement beyond chance between the generated semiquantitative interpretations and the reference labels used during evaluation. This suggests that the implemented workflow demonstrates partial analyte-level interpretation consistency despite the practical limitations associated with smartphone-based urinalysis acquisition.

Several practical factors likely influenced the obtained results, including mixed illumination conditions, glare reflections, inconsistent smartphone exposure behavior, semiquantitative overlap between adjacent analyte levels, and variability between smartphone camera systems during acquisition.

Although the obtained metrics do not yet indicate clinical-grade interpretation performance, the current implementation demonstrates the practical feasibility of integrating smartphone-based image interpretation,

semiquantitative analyte matching, and symptom-assisted screening within a structured mobile screening workflow.

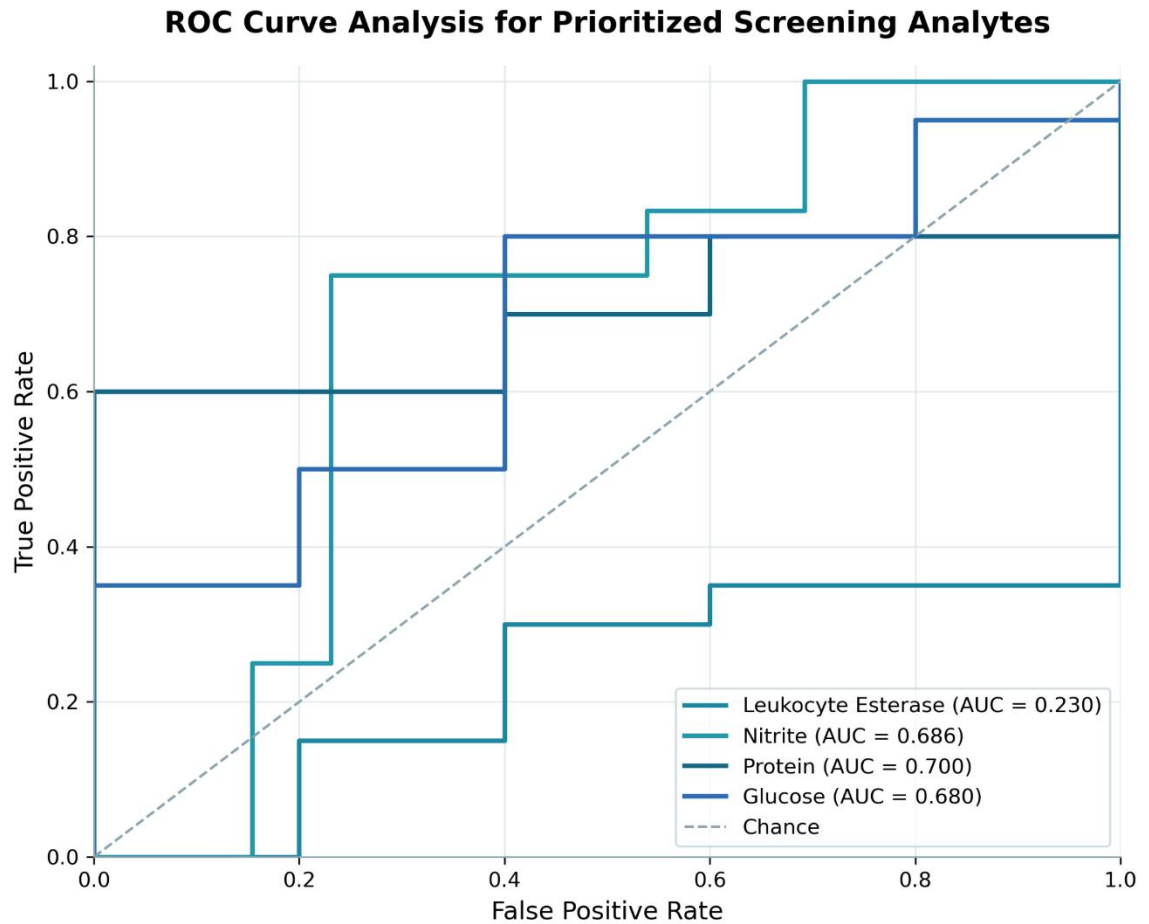


Fig. 24. ROC Curve Analysis for Prioritized Screening Analytes

The ROC analysis illustrates the discrimination capability of the implemented screening-oriented interpretation workflow across prioritized analytes associated with urinary tract infection and renal-risk screening. Protein, Nitrite, and Glucose demonstrated

moderate discrimination performance, with AUC values of 0.700, 0.686, and 0.680, respectively. In contrast, Leukocyte Esterase demonstrated comparatively weaker discrimination behavior during the current implementation-stage evaluation.

The observed variation in AUC performance across analytes may be influenced by semiquantitative overlap between reagent-strip reaction levels, illumination variability during image acquisition, and inconsistencies in smartphone-based color representation. Since the current implementation remains under ongoing dataset refinement and iterative calibration, the presented ROC behavior reflects preliminary evaluation findings rather than finalized clinical validation performance.

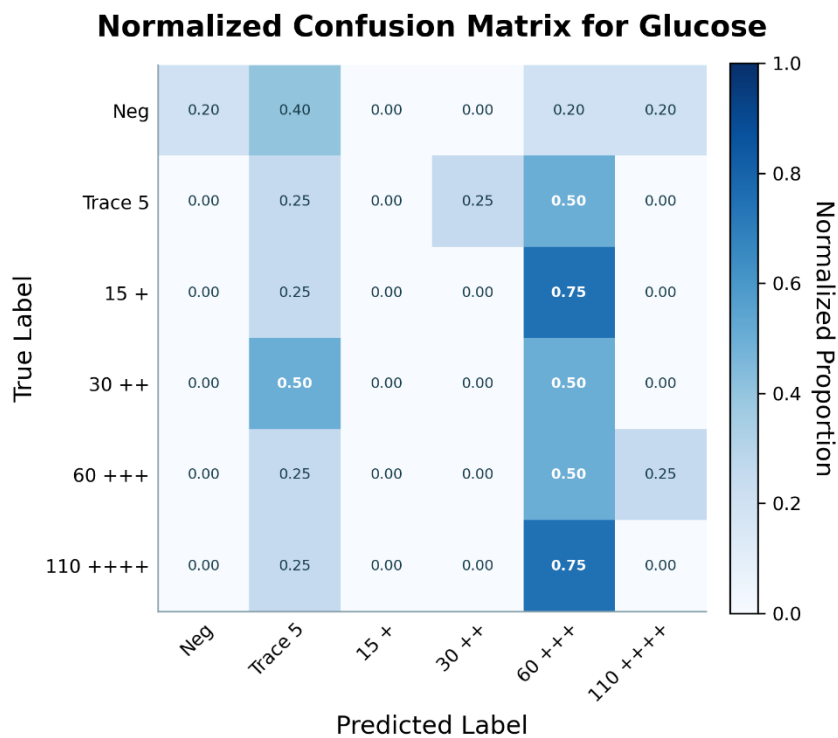


Fig. 25. Normalized Confusion Matrix for Glucose

The normalized confusion matrix for Glucose demonstrates moderate semiquantitative interpretation consistency across multiple glucose reaction levels. Several higher glucose concentration categories showed stronger proportional classification behavior, while overlap remained present between adjacent semiquantitative levels such as Trace 5, 15++, and 30++ categories. The observed behavior reflects the sensitivity of smartphone-based colorimetric interpretation to illumination conditions, exposure variation, and gradual reagent-strip color transitions.

Normalized Confusion Matrix for Leukocyte Esterase

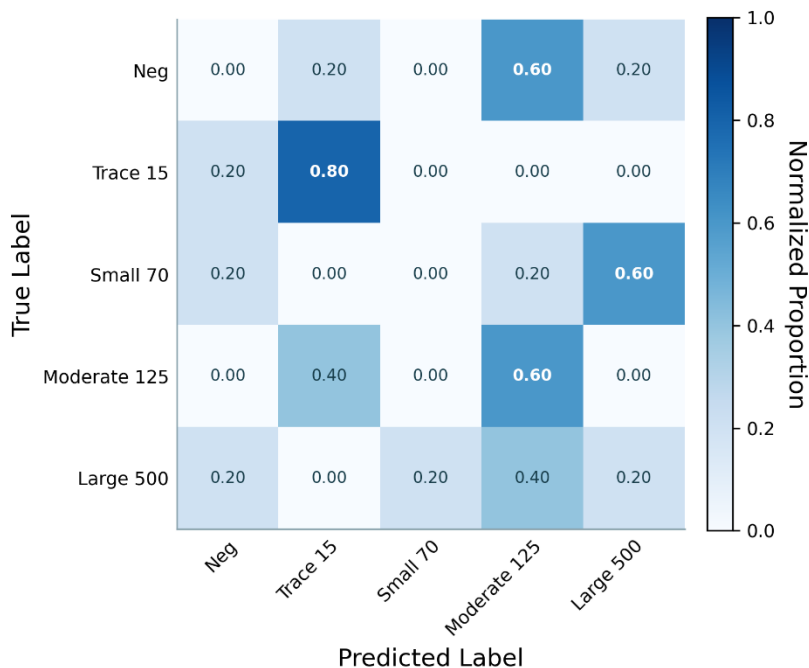


Fig. 26. Normalized Confusion Matrix for Leukocyte Esterase

The normalized confusion matrix for Leukocyte Esterase demonstrates varying interpretation consistency across semiquantitative reaction levels. Trace-level interpretations showed comparatively stronger classification behavior, while moderate overlap was observed between adjacent categories such as Moderate 125 and Large 500. The observed misclassification behavior may be influenced by gradual reagent-pad color transitions, illumination variability, and inconsistent smartphone exposure conditions during image acquisition.

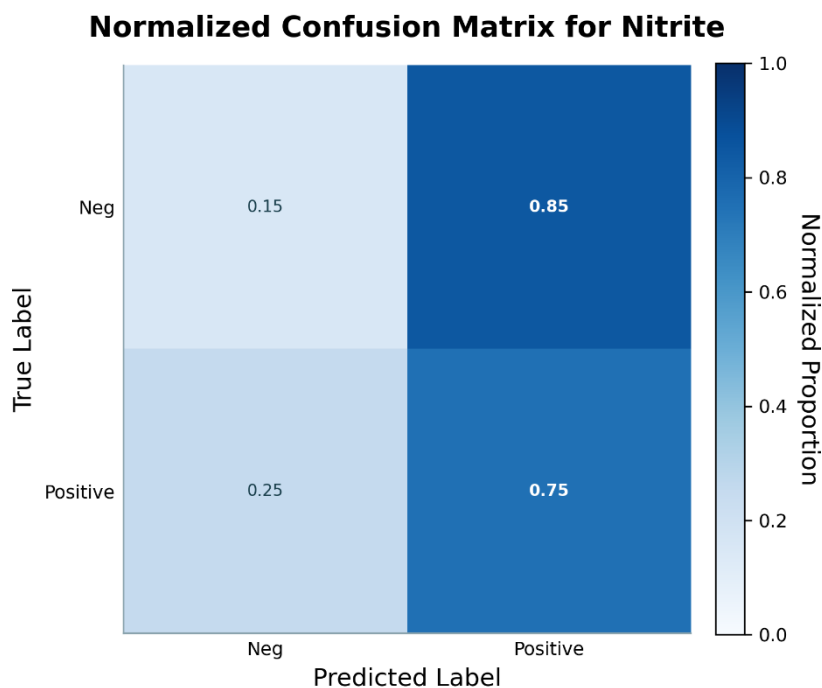


Fig. 27. Normalized Confusion Matrix for Nitrite

The Nitrite confusion matrix demonstrates comparatively stronger binary classification behavior between negative and positive reaction categories. Most positive samples were correctly classified as positive, while a portion of negative samples demonstrated overlap with positive predictions. The results suggest that Nitrite interpretation exhibited relatively stable analyte separation compared to other evaluated semiquantitative analytes within the current implementation-stage workflow.

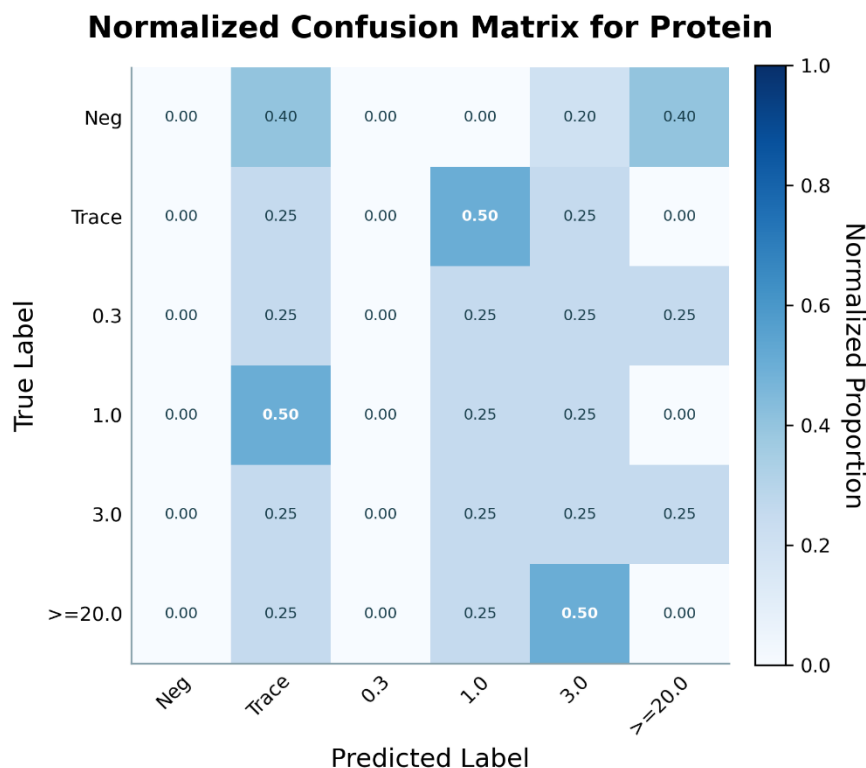


Fig. 28. Normalized Confusion Matrix for Protein

The normalized confusion matrix for Protein demonstrates moderate overlap across semiquantitative reaction levels, particularly between Trace, 1.0, and 3.0 categories. The observed behavior reflects the practical complexity of distinguishing gradual colorimetric transitions between adjacent protein concentration levels using smartphone-acquired reagent-strip images. Despite the observed overlap, several reaction levels demonstrated partial clustering behavior within the normalized matrix.

Throughout implementation and testing, several practical observations and real-world limitations were identified during smartphone-assisted urinalysis experimentation. The findings primarily involved environmental imaging variability, smartphone camera inconsistencies, and the semiquantitative complexity of reagent-strip interpretation.

TABLE XVI OBSERVED PRACTICAL CHALLENGES DURING TESTING

Observed Challenge	Effect on Interpretation
Mixed illumination	Inconsistent analyte coloration
Glare reflections	Reduced analyte visibility
Motion blur	Unstable image quality
Smartphone variability	Inconsistent color representation
Semiquantitative overlap	Ambiguous analyte interpretation

Mixed indoor lighting conditions significantly affected reagent-pad coloration and HSV feature consistency during acquisition. Variations in warm lighting, fluorescent illumination, and automatic smartphone exposure behavior occasionally altered analyte appearance during interpretation. The implemented preprocessing and localization workflow attempted to reduce portions of this variability through marker-based correction, orientation

handling, and HSV feature extraction; however, uncontrolled imaging environments continued to influence interpretation consistency during testing.

Glare reflections from wet reagent pads and unstable handheld capture additionally affected analyte visibility and image clarity. Differences in smartphone camera systems, automatic enhancement behavior, sensor characteristics, and vendor-specific camera processing also introduced inconsistencies in analyte representation across devices.

The semiquantitative nature of urine dipstick interpretation further contributed to classification ambiguity during evaluation. Several analyte categories exhibited gradual color transitions between adjacent reaction levels, making precise separation more difficult during HSV-based feature matching and k-NN interpretation. Similar overlap behavior was observed between closely related analyte categories during feature-space analysis and evaluation.

Despite these limitations, the implemented workflow demonstrated the practical feasibility of integrating smartphone-based image acquisition, analyte interpretation,

symptom-assisted screening, and structured result presentation within a unified mobile application environment. The study additionally demonstrated the potential of lightweight mobile computer vision techniques as screening-oriented clinical decision-support tools for preliminary urinalysis interpretation.

System Evaluation Results

The current Uritect implementation remains under iterative refinement and ongoing dataset expansion; therefore, comprehensive stakeholder-based system evaluation has not yet been fully conducted at the time of writing. The present implementation primarily focuses on validating the functionality of the mobile workflow, analyte interpretation pipeline, symptom-assisted screening logic, and semiquantitative interpretation behavior during development and controlled testing.

Despite the absence of a full-scale deployment evaluation, the study identified ISO/IEC 25010 as the intended software-quality evaluation framework for future system assessment. The planned evaluation will focus on software characteristics relevant to mobile healthcare screening applications, including functional suitability, usability, reliability, performance efficiency, compatibility, and portability.

The intended evaluation process will involve stakeholder-based assessment using structured evaluation questionnaires and Likert-scale scoring methods. Planned evaluators may include Medical Technology students, healthcare-related

users, faculty members, and information technology evaluators to assess both the usability and operational behavior of the implemented mobile application.

The future evaluation stage aims to assess the overall usability of the Uritect workflow, including dipstick image acquisition, semiquantitative result presentation, symptom-assisted screening interaction, scan-history management, and overall mobile user experience. Additional evaluation may also be conducted following continued dataset expansion and refinement of the analyte interpretation workflow.

At the current stage of development, the implemented system evaluation remains limited to implementation-stage functionality testing, preprocessing verification, analyte interpretation analysis, and preliminary workflow validation performed during development and experimentation.

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